Energy 356 (2026) 141223

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Adaptive coordinated frequency regulation framework and dynamic performance enhancement mechanisms for variable-speed pumped storage units with high-performance frequency support

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A R T I C L E I N F O

Keywords:
Variable-speed unit
Virtual synchronous generator
Frequency compensation
Inertia support
Dynamic stability
Primary frequency regulation
Hydro–electro-mechanical coupling

A B S T R A C T

To enhance the frequency support capability of full-size converter variable-speed pumped storage units (FSCVSPSUs) under islanded and weak-grid conditions, this study proposes a coordinated control strategy integrating an adaptive virtual synchronous generator (VSG) with frequency compensation. In this strategy, an RBF neural network is employed to realize adaptive regulation of virtual inertia, while an adaptive damping adjustment mechanism based on frequency deviation is incorporated to improve dynamic stability under frequency disturbances. In addition, a fast frequency compensation mechanism is introduced to accelerate power response during large disturbances. Leveraging the proposed control framework, a systematic analysis is conducted on the frequency regulation performance (FRP) of the unit under diverse load levels, hydraulic heads, and initial power operating conditions. This study's results corroborate the applicability of the proposed strategy significantly enhances both frequency support capability and dynamic recovery performance of FSCVSPSUs. Moreover, variations in hydraulic head and initial power exert only a minor influence on frequency response and power recovery processes, indicating a favorable decoupling between FRP and operating conditions. These findings confirm the superior frequency regulation potential of FSCVSPSUs within emerging power systems characterized by low inertia and high renewable penetration.

1. Introduction

As variable renewable energy sources (VRESs) including wind power [1] and photovoltaic generation [2] achieve ever-expanding grid integration in modern power systems, their intrinsic stochastic, volatile, and uncertain operational characteristics have introduced formidable obstacles to power grid operations. With the large-scale integration of VRESs, the proportion of synchronous generators with high rotational inertia decreases significantly, leading to the formation of low-inertia and low-damping modern power systems. Compared with traditional power systems, low-inertia power systems feature weaker frequency support capability, faster rate of change of frequency (ROCOF), larger frequency deviation under disturbances, and more vulnerable frequency security, which has become one of the core bottlenecks restricting the large-scale consumption of renewable energy. Specifically, these challenges predominantly compromise the secure and stable functioning of power systems [3]. Among all currently available large-scale energy storage technologies, pumped-storage hydropower plants (PSHPs) stand out as the most technologically mature, operationally reliable, and economically competitive option, thereby playing an indispensable role in power system regulation [4–6]. By accumulating electrical energy in periods of low grid demand, the system subsequently dispatches the stored energy to the grid during peak demand periods, PSHPs effectively perform multiple critical functions, including peak shaving and valley filling, frequency and voltage regulation, contingency reserve, and black-start capability. Particularly under high wind and photovoltaic penetration scenarios, PSHP can significantly enhance system regulation capacity and operational security margins, thereby providing essential support for the reliable large-scale incorporation of renewable energy within power systems. Accordingly, a thorough exploration of the regulatory capabilities and operational characteristics of PSHPs carries considerable theoretical implications and notable practical engineering

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https://doi.org/10.1016/j.energy.2026.141223
Received 30 January 2026; Received in revised form 1 April 2026; Accepted 30 April 2026
Available online 5 May 2026

merit. This line of research is particularly critical for underpinning the safe, stable, and high-efficiency operation of next-generation power systems [7].

In recent years, driven by continuous advancements in power electronics, pump-turbine technology, and electrical machine manufacturing, converter-based variable-speed pumped-storage technology has gradually achieved practical engineering implementation [8]. At present, research on the application of variable-speed PSHPs in power system frequency regulation remains in a stage of rapid development. As the penetration of wind and photovoltaic generation continues to rise, the share of conventional synchronous generators within the power system has undergone a remarkable contraction. This fundamental structural transformation induces a substantial reduction in system equivalent inertia, while simultaneously exacerbating the prominence of frequency stability challenges in modern power grids [9]. Under low-inertia operating conditions, the traditional frequency regulation framework dominated by thermal power and synchronous hydropower is difficult to meet the rapid and smooth control requirements of grid frequency. Therefore, developing new flexible frequency regulation resources with fast response and high regulation capacity is of great significance to maintain the frequency stability and safe operation of low-inertia power systems. Against this backdrop, variable-speed pumped-storage technology, characterized by fast power regulation capability and large-scale energy storage capacity, has attracted growing attention from both academia and industry, and is widely regarded as a key flexible resource for supporting the secure operation of future power systems [10].

At present, the control strategies based on Variable-Speed Pumped Storage Power Stations (VSPSP) are mainly divided into two categories according to the converter topology: Doubly-Fed Variable-Speed Pumped Storage Units (DFVSPSUs) and Full-Size Converter Variable-Speed Pumped Storage Units (FSCVSPSUs). For DFVSPSUs, the control system adopts a double-loop structure, including the Rotor-Side Converter (RSC) and the Grid-Side Converter (GSC). The RSC is responsible for regulating the rotor speed and active/reactive power, while the GSC is used to maintain the stability of the DC bus voltage and control the grid-side reactive power. In contrast, FSCVSPSUs connect the unit to the power grid through a full-size converter, which can independently control active and reactive power and has a more flexible control structure. Their control strategies focus on improving the dynamic response speed and tracking accuracy under transient conditions. Engineering practice has already demonstrated the considerable potential of VSPSP in frequency regulation applications. For instance, the 400 MW VSU commissioned at the Okutataragi Power Station in Japan is capable of achieving approximately 32 MW of power adjustment within 0.2 s in turbine mode, while its regulation capability can be further increased to 80 MW in pumping mode, exhibiting outstanding fast-response performance [11]. Operational statistics reveal that subsequent to the integration of VSUs, frequency regulation capability of power system has been markedly upgraded during off-peak pumping operations. For the Kansai Electric Power Company system, the likelihood of sustaining grid frequency within the (60 $\pm$ 0.1) Hz interval increased from 96.8% to 99.7%. This notable improvement has substantially enhanced the overall frequency stability and expanded the regulation margins of the system [12]. These practical examples confirm the feasibility and effectiveness of VSUs in providing rapid frequency support to power grids.

From a theoretical perspective, extensive modeling and control-oriented studies have been conducted to elucidate the frequency regulation mechanisms of VSUs. Ref. [13] established an electromagnetic–mechanical–electrical coupled model for VSUs operating in the two core operational modes of pumping and power generation, characterizing regulation behavior of unit rotational speed and active power as well as their coupling with grid frequency, thereby providing a solid theoretical foundation for frequency regulation performance (FRP) analysis. Building upon this framework, a variety of advanced control strategies have also been developed for VSPSP to improve frequency regulation performance, such as virtual inertia control (VIC), droop control, model reference adaptive control, and control based on intelligent algorithms [14–17]. These control strategies aim to meet different operational objectives, such as reducing the ROCOF, limiting frequency deviation, or balancing energy storage state and regulation performance, thereby forming a comprehensive control system for virtual synchronous power generators in frequency regulation applications. Ref. [14] introduced the concept of virtual inertia control (VIC) and explored multiple implementation strategies for VSUs. It was experimentally verified that, through the exploitation of the rotational kinetic energy stored within the combined rotors of the turbine and generator, VSUs can furnish active power support promptly at the initial stage of disturbances, with a consequent reduction in the ROCOF and a marked improvement in overall frequency stability.

To further enhance control performance, various frequency regulation strategies targeting different operational objectives have been proposed. For example, Ref. [15] combined proportional control with model reference adaptive control to fully exploit the regulation potential of VSUs, effectively limiting maximum frequency deviations while providing equivalent inertial support to power system. Meanwhile, load frequency control strategies based on advanced intelligent algorithms have attracted increasing attention [16], with studies showing that such methods can significantly improve system frequency response under varying wind power penetration levels and disturbance conditions. Furthermore, taking into account the synergistic effects of hydraulic system dynamics and unit damping characteristics with respect to frequency stability, Ref. [17] proposed a coordinated optimization strategy that balances stability and tracking performance, further validating the comprehensive advantages of PSHPs in frequency support applications.

In the field of comprehensive energy frequency regulation, scholars have conducted extensive research on the coordinated control of wind-storage combined systems. For example, Ref. [18] proposes a control strategy for wind energy and energy storage to participate in the primary frequency regulation of power systems, considering an energy storage recovery strategy. During the primary frequency regulation process, the combined output of wind turbines adopting virtual inertia control and energy storage batteries adopting droop control can effectively suppress the system frequency drop; during the frequency regulation period, the energy storage batteries are charged through a recovery strategy to ensure that the state of charge (SOC) of the energy storage batteries is in the state of maximum frequency regulation capacity. However, most of these studies focus on electrochemical energy storage with small capacity and short duration, while research on large-scale and long-duration energy storage represented by pumped storage is relatively insufficient, especially the coordinated frequency regulation mechanism with full-size variable frequency units still needs further research. In terms of the choice of practical-oriented control strategies, impact of different control priorities on FRP has also been investigated. Comparative engineering studies indicate that, when VSU units provide ancillary services to the grid, power-priority control strategies generally yield more favorable regulation performance [19]. At the same time, frequency response control approaches based on speed-priority concepts have continued to evolve. By introducing additional speed correction terms and restructuring the control relationship between rotor speed and grid frequency, these methods enable rapid coordinated regulation of unit speed and active power under frequency disturbances [20]. Further system-level analyses have examined the influence of VSU participation in frequency regulation on grid frequency stability, confirming the effectiveness of the associated control strategies in providing frequency support [21]. Notably, to address the difficulty of directly responding to system frequency deviations under decoupled control conditions in doubly fed VSUs, Ref. [7] introduced an additional guide vane opening control loop, allowing frequency deviation signals to simultaneously act on both generator power control and pump-turbine guide vane control. The results demonstrated that this approach can effectively mitigate the maximum post-disturbance frequency deviation. Meanwhile, it diminishes the steady-state primary frequency regulation error under both generating and pumping modes, thereby expanding the implementation pathways for frequency support utilizing VSUs.

It is crucial to compare the frequency regulation performance between fixed-speed pumped-storage units (FSPSUs) and VSPSPs, as their differences in controller structure, control strategy, and regulation characteristics directly affect their application effects in low-inertia power systems. In terms of controller structure, FSPSUs adopt a synchronous generator directly connected to the grid, with a simple control structure that mainly relies on the governor to adjust guide vane opening, thereby regulating unit output. However, this structure lacks flexible power control capability and is limited by the mechanical response speed of the governor. In contrast, VSPSPs are equipped with power electronic converters, which enable decoupled control of active and reactive power. Although their control structure is more complex, it provides more flexible regulation capabilities. In terms of control strategy, FSPSUs mainly adopt proportional-integral (PI) control based on frequency deviation, focusing on steady-state frequency regulation, but their dynamic response is slow, making it difficult to adapt to the rapid frequency fluctuations in low-inertia power systems. In contrast, VSPSPs integrate VIC, droop control, and other advanced strategies, which can utilize the rotational kinetic energy of the unit to provide rapid inertial support in the initial stage of frequency disturbance and adjust active power output in real time according to frequency changes, achieving both dynamic and steady-state frequency regulation. In terms of regulation characteristics, FSPSUs have a slow response speed, a limited regulation range, and poor adaptability to transient disturbances. In contrast, VSPSPs have a fast response speed, a wide regulation range, and strong ability to suppress frequency fluctuations, which can better meet the frequency regulation requirements of low-inertia power systems with high renewable energy penetration.

Overall, existing studies generally acknowledge the promising application prospects of VSUs in power system frequency regulation, with participation mechanisms mainly based on frequency droop control and VIC. However, in terms of research scope and methodology, most existing studies focus primarily on doubly fed variable-speed units, while systematic investigations into FSCVSPSUs remain relatively limited. Moreover, in low-inertia power systems with high renewable penetration, the adaptive adjustment mechanism of virtual inertia parameters considering disturbance magnitude and operation conditions, as well as the coordinated optimization between unit damping characteristics and fast power compensation capability under transient disturbances, still lack in-depth theoretical analysis and verification. Further in-depth research is therefore required to fully exploit the frequency support potential of FSCVSPSUs.

Leveraging the findings derived from the preceding analyses, the salient contributions of the current research are enumerated as follows.

- An adaptive frequency regulation simulation model has been proposed for full-size converter variable-speed unit.
- Propose a frequency regulation strategy to enhance unit support capability and stability.
- Uncover evolution mechanism of frequency response and hydraulic characteristics in various scenarios.
- Reveal significant decoupling between frequency regulation indicators and conditions.

The rest of this manuscript is structured as follows. Section 2 formulates a refined mathematical model of FSCVSPSUs, providing a theoretical basis for subsequent analytical work. Section 3 presents a frequency-oriented control strategy derived from the aforementioned model. Section 4 evaluates and analyzes the frequency dynamic response characteristics and hydraulic transient behaviors of the units under various typical operating scenarios. Lastly, a comprehensive summary of the key findings, alongside the principal conclusions derived from this investigation, is presented in Section 5.

2. High-fidelity modeling of FSCVSPSUs

A PSHP can be regarded as a time-varying, strongly nonlinear hydro–mechanical–electrical coupled system, which mainly consists of pressurized water conveyance system, pump–turbine, the governing system, generator/motor, and the excitation system [22]. These subsystems are mutually coupled through hydraulic transients, mechanical rotation, and electromagnetic processes, thereby jointly determining response characteristics and stability margins of unit.

Compared with conventional units, a FSCVSPSU [23] is connected to power grid via full-size converter (FSC) rated at same capacity as the generator, enabling adjustable rotational speed while maintaining constant grid frequency. This configuration significantly enhances regulation flexibility and energy conversion efficiency. The grid-connected structure of the FSCVSPSU is illustrated in Fig. 1.

In this study, a modular modeling approach is adopted. Mathematical models are developed separately for the pressurized water conveyance system, the pump–turbine, governing, generator/motor, excitation system, and control strategy of the FSC, thereby providing a unified foundation for subsequent coupled analysis and control design.

2.1. Water conveyance system model

The pressurized water delivery system of a conventional PSHP is generally composed of core components, including upper and lower reservoirs, surge tanks, penstocks, bifurcation pipes, spiral casings, and draft tubes. Such structural components exert a pivotal influence on the formation of the system's hydraulic response in the course of unit operation.

In regard to the transient hydraulic phenomena resulting from significant variations in operating conditions of pumped storage hydropower plants (PSHPs), it is imperative to develop a dynamic mathematical model for the penstock system via the method of characteristics (MOC) [24], so as to precisely characterize the temporal variations of pressure and flow within the system. The mathematical model describing one-dimensional unsteady flow in the pressurized water conveyance system is formulated as follows:

$$\begin{cases} \dfrac{\partial H(x,t)}{\partial x} + \dfrac{1}{gA}\dfrac{\partial Q(x,t)}{\partial t} + \dfrac{f|Q(x,t)|}{2gDA^2}Q(x,t) = 0 \\ \dfrac{a^2}{gA}\dfrac{\partial Q(x,t)}{\partial x} + \dfrac{\partial H(x,t)}{\partial t} = 0 \end{cases} \tag{1}$$

where H is hydraulic head in penstock; Q denotes flow rate in pipeline; x represents spatial coordinate along the pipeline; t denotes time; f is Darcy–Weisbach friction factor; D denotes pipeline diameter; and a represents the water-hammer wave speed.

When the MOC is employed to tackle the partial differential equations specified in Eq. (1), it is generally necessary to construct ordinary differential equations along the characteristic lines, thereby significantly reducing the computational complexity of the problem. Accordingly, the partial differential equations in Eq. (1) can be transformed into:

$$\begin{cases} \dfrac{dH}{dt} \pm \dfrac{a}{gA}\dfrac{dQ}{dt} \pm \dfrac{af}{2gDA^2}|Q|Q = 0 \\ \dfrac{dx}{dt} = \pm a \end{cases} \tag{2}$$

Based on finite difference method, a characteristic grid is constructed to discretize Eq. (2), through which flow rate and hydraulic head in the pipeline at any given time can be solved.

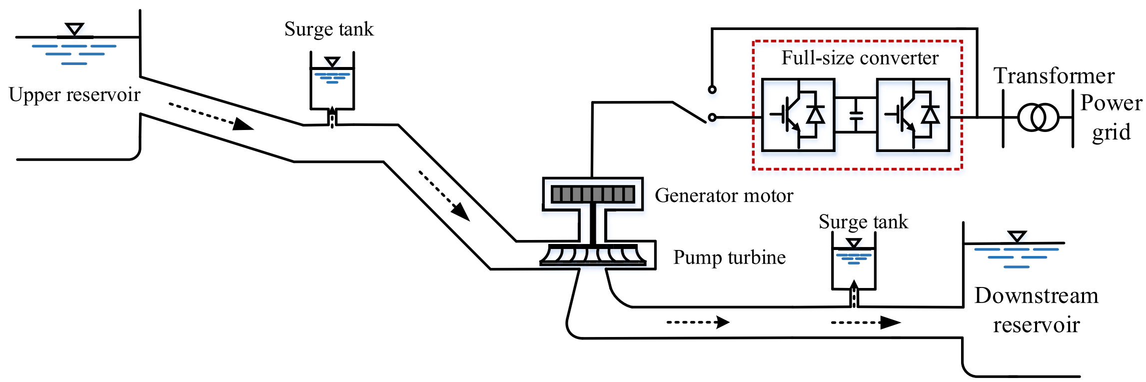

Fig. 1. Grid connection diagram of FSCVSPSU.

2.2. Pump-turbine model

A pump–turbine is a type of hydraulic machine with bidirectional operating capability, allowing rapid switching between generating mode and pumping mode, and exhibiting pronounced nonlinear and bimodal operating characteristics. The purpose of mathematical modeling is to characterize the dynamic response relationships among key hydro–mechanical variables—including rotational speed, flow rate, hydraulic head, and mechanical torque—with respect to variations in guide vane opening.

Depending on specific research objectives, existing mathematical models of pump–turbines can be categorized into linearized models, internal characteristic models, and full characteristic curve models based on experimental data. For the purposes of this research, a full characteristic curve model was utilized as the foundational analytical framework. However, full characteristic curves of pump–turbines typically contain an inverse S-shaped characteristic region, within which severe phenomena such as intersection, clustering, and twisting of guide vane opening curves occur. These features may lead to multi-valued solutions in numerical simulations.

To address such multi-valued problems and guarantee numerical robustness, an improved Suter transformation (IST) approach is adopted in this study for processing the complete characteristic curves of pump-turbines. The mathematical formulation of IST is presented as follows [25]:

$$\begin{cases} WH(x,y) = \dfrac{h(y+C_y)^2}{n^2+q^2+C_h h} \\ WM(x,y) = \dfrac{(m+k_1 h)(y+C_y)^2}{n^2+q^2+C_h h} \\ x = \arctan\left[\left(q+k_2\sqrt{h}\right)/n\right], n \geq 0 \\ x = \pi + \arctan\left[\left(q+k_2\sqrt{h}\right)/n\right], n < 0 \end{cases} \tag{3}$$

where *WH(x,y)* and *WM(x,y)* denote transformed variables describing full characteristic curves after the IST, x is independent variable; h, y, n, q, and m represent relative values of hydraulic head, guide vane opening, rotational speed, flow rate, torque, respectively; $k_1 > |M_{11\ \max}|/M_{11r}$; $k_2 = 0.5 \sim 1.2$; $C_y = 0.1 \sim 0.3$; $C_h = 0.4 \sim 0.6$. Fig. 2 depicts the full characteristic curves of the pump–turbine acquired through the transformation specified in Eq. (3).

2.3. Governor model

Unlike traditional hydropower units, PSHPs possess distinct bidirectional operational features, which necessitates that the governor delivers efficient regulatory performance under both generating and pumping modes. In current engineering applications, governing system of PSHPs mainly employs classical proportional–integral–derivative (PID) control strategy.

Through the combined actions of proportional, integral, and derivative terms, the PID controller enables real-time response to control deviations, cumulative elimination of steady-state errors, and dynamic trend prediction, thereby effectively enhancing system stability and regulation performance. The model of governor is formulated as follows:

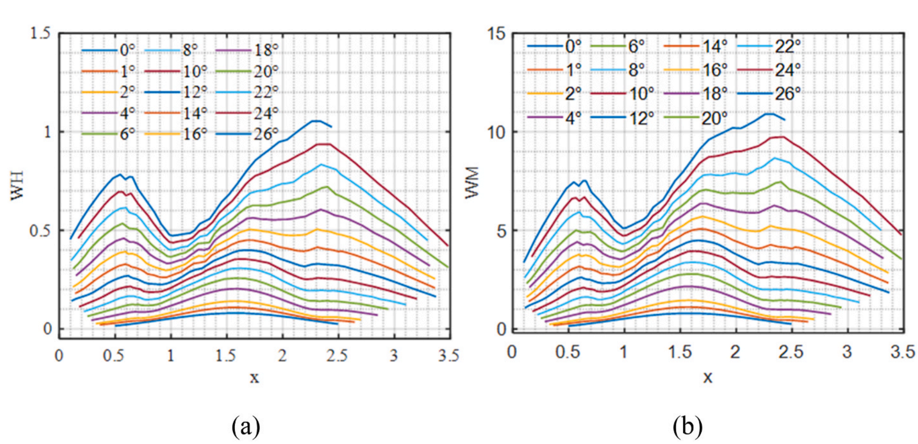

Fig. 2. The complete characteristic curve of the pump-turbine following Suter transformation has been enhanced: (a)WH; (b)WM.

$$u = \left(K_p + \frac{Ki}{s} + K_d s\right)(n_{ref} - n) \tag{4}$$

Where K_p, K_i, and K_d denote PID parameters, respectively; s is Laplace transform operator; u represents output signal of PID.

In the governing system of a PSHP, in addition to the PID controller, the actuator constitutes an indispensable key component. Its mathematical formulation is given as follows [26]:

$$\frac{\mathrm{dy}}{\mathrm{dt}} = \frac{1}{T_y}(u - y) \tag{5}$$

Where T_y denotes the servomotor's response time.

2.4. Synchronous generator model

The operating mode of a synchronous reversible motor–generator is governed by the direction of mechanical power. When it is positive, the unit operates in generating mode; when negative, the unit transitions to motoring mode. To accurately characterize the electromagnetic and mechanical dynamic behaviors, the synchronous machine model established in this study integrates voltage equations, flux linkage equations, and the rotational motion equation, which enables comprehensive description of the unit's dynamic performance under both operating modes [27]:

$$\begin{cases} u_d = \dfrac{\mathrm{d}\psi_d}{\mathrm{dt}} - r_s i_d - \omega\psi_q \\ u_q = \dfrac{\mathrm{d}\psi_q}{\mathrm{dt}} - r_s i_q + \omega\psi_d \\ u_f = \dfrac{\mathrm{d}\psi_f}{\mathrm{dt}} + r_f i_f \end{cases} \tag{6}$$

$$\begin{cases} \psi_d = -L_d i_d + m_{af} i_f \\ \psi_q = -L_q i_q \\ \psi_f = -\dfrac{3}{2} m_{af} i_d + L_f i_f \end{cases} \tag{7}$$

$$\begin{cases} \dfrac{\mathrm{d}\omega}{\mathrm{dt}} = \dfrac{M_t - \left(\psi_d i_q - \psi_q i_d\right) - D(\omega - 1)}{T_a} \\ \dfrac{\mathrm{d}\delta}{\mathrm{dt}} = \omega_0(\omega - 1) \end{cases} \tag{8}$$

Where u_d and u_q are voltage components in the dq reference frame; ψ_d and ψ_q denote magnetic flux linkages in the dq reference frame; i_d and i_q are corresponding current components; r_s is the stator resistance; ω denotes the angular speed of the machine; u_f is the excitation voltage;f ψ_f represents the excitation flux linkage; i_f denotes the excitation current; L_f denotes the excitation winding inductance; M_t is output torque of pump–turbine; T_a is the inertia time of unit; D is damping coefficient of synchronous generator; δ denotes the rotor electrical power angle; and ω_0 is the rated angular speed.

2.5. Control strategy of the FSC

For a FSCVSPSU, the FSC is applied to the reversible PSHP, primarily comprising a machine-side converter (MSC) [28] and a grid-side converter (GSC) [29]. Directly interfacing with the synchronous machine, the MSC is responsible for regulating the stator current, as well as the generator's electromagnetic torque and power. In contrast, the GSC exchanges energy with the power grid to maintain DC-link voltage stability and grid-side current stability. Coupled via a common DC-link, these two converters make DC-link voltage regulation a decisive factor for energy flow and power balance within the entire converter system.

During the operation of the FSC, one of the primary control objectives of MSC is to maintain DC-link voltage close to its reference value, thereby ensuring real-time power balance between the MSC and the GSC during energy transmission. In the control architecture, the DC-link voltage is typically selected as the outer-loop control variable of the MSC, forming a voltage regulation loop aimed at minimizing the steady-state DC voltage deviation. This outer loop measures the deviation between the actual DC-link voltage and its reference value and generates a current reference signal through a controller incorporating proportional and integral actions. The generated reference is subsequently input to the inner current regulation loop of the MSC, in which the d-axis current is adjusted to meet the required electromagnetic power. This approach enables the stable operation of the DC-link voltage.

Accordingly, the mathematical model of the DC-link voltage outer-loop control for MSC can generally be expressed as follows:

$$\begin{cases} i_d^* = \left(K_p + \dfrac{K_i}{s}\right)\left(V_{dc}^* - V_{dc}\right) \\ i_q^* = \left(K_p + \dfrac{K_i}{s}\right)\left(Q_{dc}^* - Q_{dc}\right) \end{cases} \tag{9}$$

Where i_d^* and i_q^* denote the reference values of machine-side current control loop; V_{dc}^* and V_{dc} represent the reference and measured DC-link voltages; and Q_{dc}^* and Q_{dc} denote the reference and measured reactive power on the machine side, respectively.

The inner-loop control of the MSC is generally realized through a current-controlled configuration, with its core goal being the achievement of rapid and precise adjustment of the synchronous machine's stator currents. This ensures the timely and stable electromagnetic power response. Such a response is maintained for the unit across various operating conditions.

To enhance control performance, a feedforward decoupling strategy is commonly adopted in the MSC current inner loop. By compensating the cross-coupling terms through feedforward control, the dq-axis current equations can be approximately decoupled, allowing independent regulation of active and reactive power channels, respectively.

On the basis of the decoupled current dynamics, proportional–integral (PI) controllers are employed in the inner loop to regulate the error between the reference currents and the measured currents, thereby ensuring satisfactory tracking performance and steady-state accuracy. The outputs of PI serve as the voltage reference commands for the FSC modulation stage. Voltage commands of this sort are imposed on the MSC via modulation methods including sinusoidal pulse-width modulation (SPWM) and space vector pulse-width modulation (SVPWM), thereby achieving real-time regulation of electromagnetic quantities at the machine terminals.

Considering the above control structure, the mathematical model of the MSC current inner-loop control can be expressed as:

$$\begin{cases} u_d = -\left(K_{pi} + \dfrac{K_{ii}}{s}\right)\left(i_d^* - i_d\right) + \omega L i_q + e_d \\ u_q = -\left(K_{pi} + \dfrac{K_{ii}}{s}\right)\left(i_q^* - i_q\right) + \omega L i_d + e_q \end{cases} \tag{10}$$

Where u_d and u_q denote the dq axis components of the MSC-side AC voltage; L represents the filter inductance on the machine side; and e_d and e_q are the dq axis components of the terminal voltage of the synchronous machine. An overall block diagram for the MSC control strategy is presented in Fig. 3.

The GSC is not only tasked with delivering or absorbing energy to/from the power grid, but also accountable for ensuring DC-link voltage stability, achieving reactive power regulation, guaranteeing grid current quality, and facilitating coordinated interaction between the unit and the grid across various operating scenarios.

The grid-side control system is typically configured with a dual-loop structure: an outer voltage control loop and an inner current control loop. Taking voltage as the reference variable, the outer voltage loop

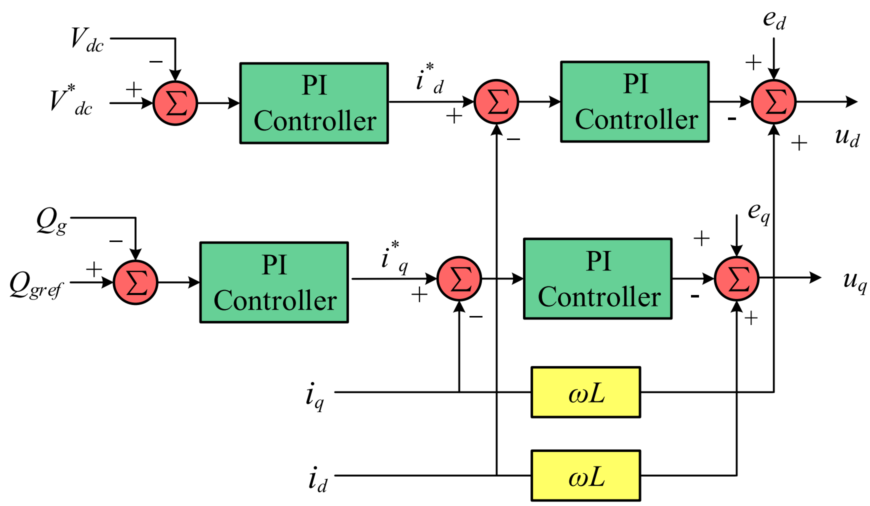

Fig. 3. Control block diagram of the MSC.

regulates the converter output voltage to ensure accurate reproduction of the desired voltage characteristics. Specifically, the deviation between the reference and measured output voltages is computed; after processing by a PI controller, this deviation generates current reference signals in the d-q reference frame. These current commands are subsequently tracked by the inner current loop, enabling the converter to rapidly respond to the voltage regulation demands from the outer loop.

Specifically, the PI of outer voltage loop processes the voltage deviation, yielding the reference values for dq-axis currents. The relevant mathematical expression is summarized as follows:

$$\begin{cases} i^*_{dref} = \left(K_{pu} + \dfrac{K_{iu}}{s}\right)(u_{dref} - u_d) - \omega C_f e_{oq} + i_{sd} \\ i^*_{qref} = \left(K_{pu} + \dfrac{K_{iu}}{s}\right)(u_{qref} - u_q) + \omega C_f e_{od} + i_{sq} \end{cases} \tag{11}$$

Where i^*_{dref} and i^*_{qref} are the reference values of the current inner control loop; u_{dref} and u_{qref} represent the reference values of the voltage outer loop in the dq frame; C_f is grid-side filter capacitance; and e_{od} and e_{oq} are the dq axis components of grid-side output voltage of converter.

The mathematical model corresponding to the current inner control loop may be formulated as follows:

$$\begin{cases} v_d = \left(K_{pi} + \dfrac{K_{ii}}{s}\right)\left(i^*_{dref} - i_d\right) - \omega L_f i_q + e_{od} \\ v_q = \left(K_{pi} + \dfrac{K_{ii}}{s}\right)\left(i^*_{qref} - i_q\right) + \omega L_f i_d + e_{oq} \end{cases} \tag{12}$$

Where v_d and v_q are the reference input voltages for the PWM modulation strategy; and L_f represents the grid-side filter inductance. Fig. 4 illustrates the voltage–current dual closed-loop control strategy block diagram for the FSC's GSC.

3. Frequency regulation strategies for FSCVSPSUs

FSCVSPSUs connect to the power grid via power electronic converters. Consequently, they intrinsically lack mechanical inertia and damping, giving rise to dynamic characteristics dominated by low inertia and weak damping. Under such circumstances, the power system's frequency stability and its primary frequency regulation capability deteriorate remarkably when exposed to disturbances, such as sudden load surges or fluctuations in wind and photovoltaic power generation.

To address this challenge, the VSG [30] concept has been introduced. By embedding the rotor dynamic equations and electromagnetic energy balance mechanisms of conventional synchronous generators into inverter control strategies, the VSG enables power electronic devices to emulate virtual inertia and virtual damping. In this manner, the inertia

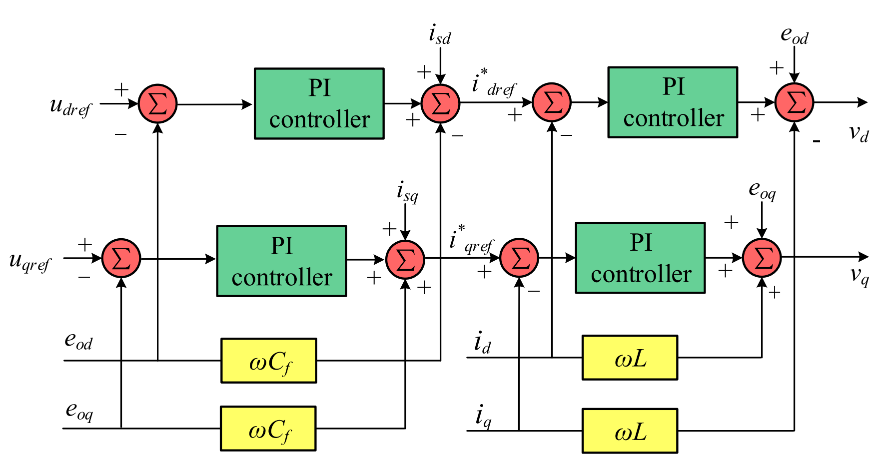

Fig. 4. Control block diagram of the GSC.

support and frequency response characteristics of traditional power systems can be partially restored, thereby enhancing system frequency stability and dynamic regulation performance.

In conventional VSG control frameworks, the inertia coefficient J and damping coefficient D are typically designed as fixed parameters. However, fixed-parameter control strategies are intrinsically constrained: concurrent attainment of rapid dynamic responsiveness and desirable steady-state stability proves unattainable under fluctuating operational conditions. To overcome these limitations, adaptive control strategies have been introduced into the VSG control paradigm. The core idea of these strategies is to dynamically adjust the key control parameters J and D in accordance with real-time system frequency conditions, thereby achieving effective parameter–state matching.

In this study, frequency deviation and its rate-of-change are adopted as inputs for the adaptive VSG controller, allowing online tuning of inertia and damping coefficients. This adaptive mechanism achieves a balanced compromise between suppressing the frequency change rate and mitigating steady-state frequency deviation, thus improving the FRP in both transient and steady-state stages.

From a mechanistic perspective, the transient frequency response of adaptive VSG control is jointly governed by the J and D. The inertia coefficient J primarily affects the ROCOF: a higher value of J enables effective mitigation of rapid frequency deviations in the power system, but at the expense of a slower system response and a prolonged recovery time. In contrast, the damping parameter D mainly regulates frequency deviation; increasing D reduces steady-state frequency error, yet diminishes response sensitivity.

By introducing a frequency-dynamic–based parameter self-adjustment mechanism, the adaptive VSG control strategy overcomes the inherent trade-off between dynamic flexibility and stability that characterizes conventional fixed-parameter control schemes. Fundamentally, it establishes dynamic coordination between inertia and damping, enabling FSCVSPSUs with adaptive VSG control to not only deliver effective equivalent inertia support but also actively engage in transient stability control during primary frequency regulation. The proposed approach markedly improves the power system's dynamic response performance and frequency stability margin under sudden load disturbances.

3.1. The principle of radial basis function neural network

The Radial Basis Function (RBF) neural network is a typical three-layer feedforward neural network, consisting of an input layer, a hidden layer, and an output layer. There are only one-way connections between neurons in each layer, with no feedback loops, which conforms to the core characteristics of multi-layer feedforward networks. Among them, the input layer is responsible for receiving external input signals and directly transmitting them to the hidden layer; the hidden layer neurons adopt radial basis functions as activation functions, whose core role is to perform nonlinear mapping on the input signals, convert the signals in the input space to a high-dimensional feature space, and achieve accurate fitting of complex nonlinear relationships; the output layer performs a linear weighted sum of the mapping results of the hidden layer and outputs the final control compensation signal. The core advantage of this network lies in the local approximation characteristics of the radial basis functions in the hidden layer, which endow it with the characteristics of fast learning speed, high fitting accuracy, and strong generalization ability. It can realize effective control of nonlinear systems without complex network training iterations. The structure of the RBF neural network is shown in Fig. 5.

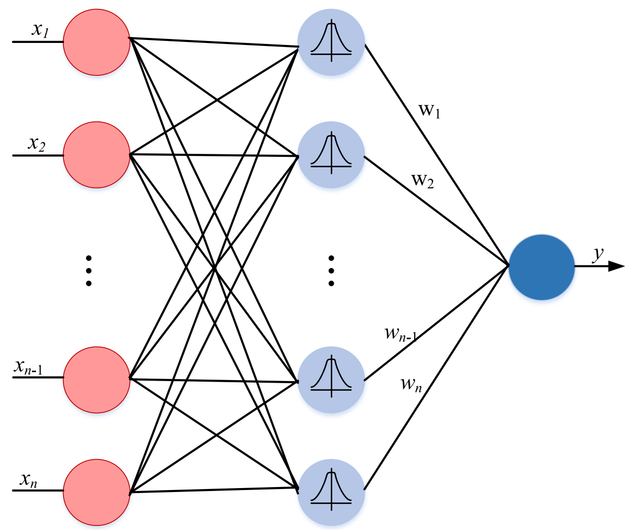

Fig. 5. Structure diagram of RBF neural network.

3.2. Adaptive VSG control strategy

The previous section described the principle of the RBF neural network. To realize the adaptive adjustment of VSG control parameters based on real-time measurement, this section proposes a real-time dynamic adjustment method for the virtual inertia J based on the RBF neural network [31]. Fig. 6 shows a schematic diagram of the adaptive optimization of VSG parameters based on the RBF neural network in energy storage applications. As shown in Fig. 6, the RBF neural network used in this study is configured as a three-layer feedforward structure, including an input layer, a hidden layer, and an output layer [32]. Specifically, the input variables of the network include the frequency deviation and its rate of change, while the output corresponds to the adaptive virtual inertia J. The hidden layer adopts the Gaussian basis function to approximate the nonlinear mapping between the system state and the control parameters. In this study, the number of hidden layer nodes is selected as 5, which is determined based on a comprehensive trade-off between approximation accuracy and computational complexity. Specifically, too few hidden layer nodes may lead to insufficient nonlinear approximation capability, making it impossible to capture the complex nonlinear relationship between the system state and the VSG control parameters. On the contrary, too many hidden layer nodes will significantly increase the computational burden of the network, which may impair the real-time performance of VSG parameter adjustment. The rotor angular velocity ω and $d\omega/dt$ are selected as the inputs of the RBF neural network, while the virtual moment of inertia J is taken as the network output. The input function of the RBF neural network is expressed as follows:

$$O_n^{(1)} = x(i), i = 1, 2 \tag{13}$$

The input function corresponding to the hidden layer of RBF neural networks is expressed as:

$$net_m^{(2)} = x, x = O_n^{(1)} \tag{14}$$

The output mapping function of the hidden layer in the RBF neural network is formulated as follows:

$$O_m^{(2)} = \exp\left(-\frac{\|y - c_m\|^2}{2b_m^2}\right), m = 1, 2, ..., 5 \tag{15}$$

Here, y denotes the input to the hidden layer, c_m represents the center of the m -th hidden-layer neuron, and b_m denotes the width (spread) parameter of the Gaussian basis function.

The input supplied to the output layer of the RBF neural network is determined by:

$$net_l^{(3)} = \sum_{m=1}^{5} w_m^{(3)} O_m^{(2)} \tag{16}$$

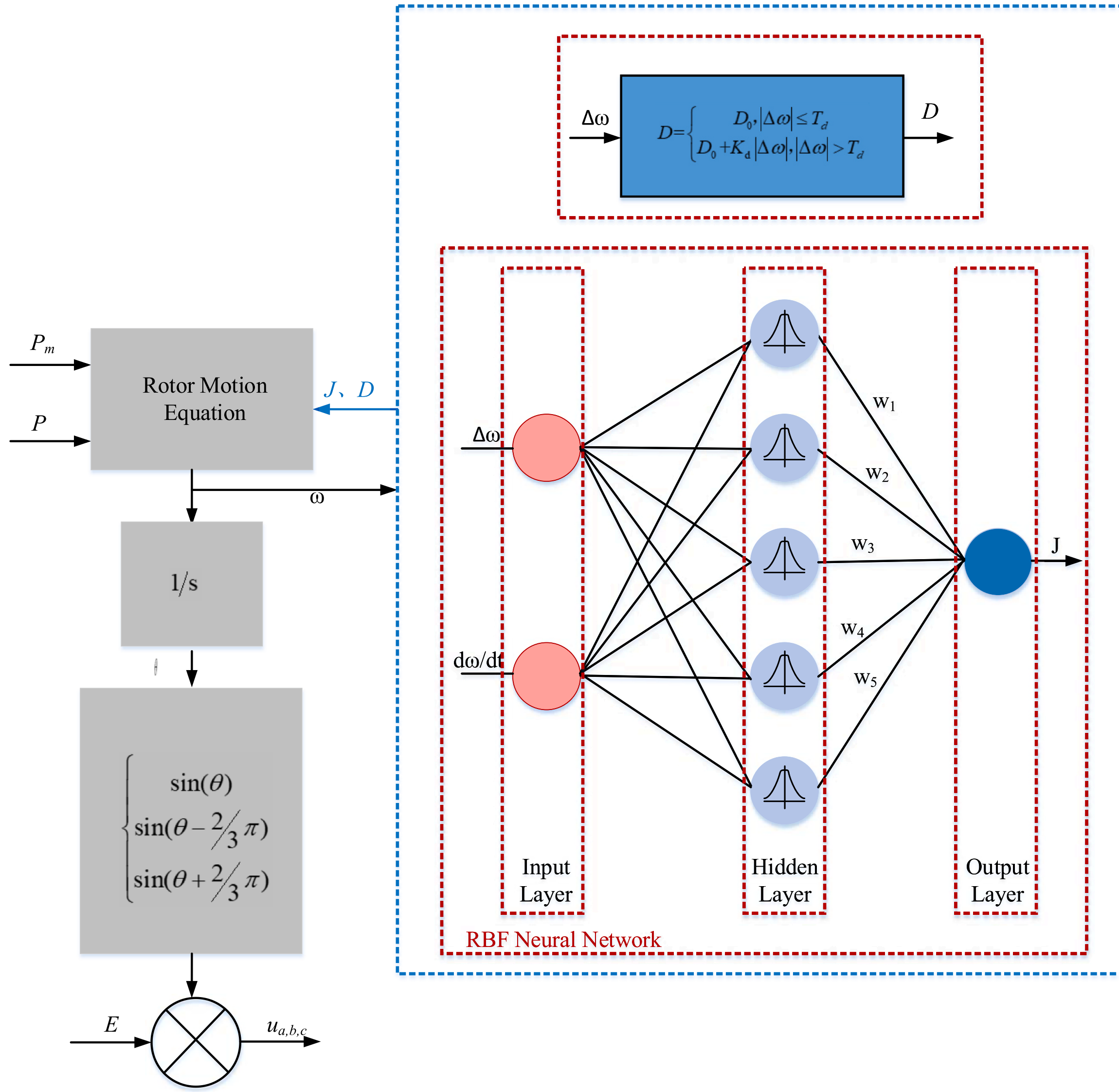

Fig. 6. Schematic diagram of adaptive parameter optimization for the VSG.

Where, $w_m^{(3)}$ denotes the weight of the RBF neural network.

The output of the RBF neural network output layer is expressed as follows:

$$J = O_l^{(3)} = \frac{ue^z}{e^z + e^{-z}} \tag{17}$$

Here, u denotes the upper bound of the virtual moment of inertia J, and z represents the input to the output layer.

The performance function of the RBF neural network is formulated as follows:

$$E(k) = \frac{1}{2}(\omega_0 - \omega)^2 \tag{18}$$

In this study, a gradient descent algorithm with an inertia (momentum) term was employed to update the weights of the hidden layer. This approach was chosen because it is simple, has low computational complexity, and is suitable for real-time applications. Compared with more complex optimization algorithms, when an appropriate learning rate is selected, gradient descent can achieve efficient online parameter adjustment and has stable convergence characteristics. The corresponding weight update formula is as follows:

$$\Delta\omega_m^{(3)}(k) = -\beta\frac{\partial E}{\partial \omega_m^{(3)}} + \alpha\Delta\omega_m^{(3)}(k-1) \tag{19}$$

Here, β and α denote the learning rate and the inertia (momentum) coefficient of the RBF neural network, respectively.

The adaptive control strategy for the damping coefficient D is formulated as follows:

$$D = \begin{cases} D_0, |\Delta\omega| \le T_d \\ D_0 + K_d|\Delta\omega|, |\Delta\omega| > T_d \end{cases} \tag{20}$$

Here, D_0 denotes the damping coefficient of the virtual synchronous generator under steady-state operating conditions, K_d is the regulation gain of the damping coefficient D, and T_d represents the variation threshold. The value of T_d in the text is 0.05.

3.3. Frequency compensation control strategy

FSCVSPSUs' rotational speed decouples from grid frequency, which inherently restricts their capacity to actively deliver frequency support to the power system. Consequently, an auxiliary frequency compensation control (FCC) strategy is indispensable to facilitate FSCVSPSUs' participation in grid frequency regulation. Specifically, under the power-priority operating mode, the frequency support capability can be achieved by incorporating a frequency compensation loop into the adaptive VSG control of the GSC.

Specifically, the FCC modifies active power reference of adaptive VSG controller according to the grid frequency deviation, thereby regulating the capability of the unit to absorb or release rotational kinetic energy and provide effective frequency support. During frequency regulation, sustained power support must still be ensured via regulating the guide vane opening to modulate the pump-turbine output. Correspondingly, the synchronous generator's electromagnetic power is adopted as the input for speed optimization, and a coordinated primary frequency regulation control strategy integrating the pump-turbine subsystem and the FSC is proposed.

Drawing on the aforementioned control philosophy, the control block diagram for FSCVSPSUs engaging in primary frequency regulation under the power-priority operation is presented in Fig. 7.

where the compensation power generated by the FCC is expressed as follows:

$$\Delta P = K_p\left(f_{ref} - f\right) \tag{21}$$

$$K_p = \begin{cases} 1 + |\Delta f|, \Delta f > M_d \\ 1, \Delta f \le M_d \end{cases} \tag{22}$$

Here, ΔP denotes the compensation power generated by the FCC, K_p is the primary frequency regulation gain of the FCC, f_{ref} represents the reference grid frequency, f is the real-time grid frequency, and Δf denotes the grid frequency deviation.

The power constraint is expressed as follows:

$$P_{ref} = \Delta P + P^*, P_{ref} \in [P_{\min}, P_{\max}] \tag{23}$$

Here, P_{ref} denotes the power reference of the FSCVSPSU, P^* represents the dispatch command issued to the FSCVSPSU, and $P_{\min}$ and $P_{\max}$ are the minimum and maximum power limits of the FSCVSPSU, respectively.

4. Results and analysis

4.1. The influence of the control strategy of FSCVSPSU on frequency regulation

This section focuses on the operational characteristics of the FSCVSPSU under turbine mode. By connecting the FSCVSPSU to an independent infinite bus system, fast power response simulation tests are carried out under the coordinated control strategy of adaptive virtual moment of inertia, damping coefficient and frequency compensation to complete the comprehensive evaluation of the unit's operational characteristics. Table 1 presents some key parameters of the FSCVSPSU.

4.1.1. Analysis of system frequency response under different parameters

To evaluate the effects of virtual moment of inertia J and damping coefficient D on the system frequency regulation performance, three sets of different parameter values are selected respectively in this paper to carry out comparative simulation studies. At the simulation time $t =$ 100s, a load disturbance with an amplitude of 0.05p.u. of the rated power is suddenly applied. The dynamic frequency response characteristics of the system when the FSCVSPSU participates in the primary frequency regulation of the power grid under different parameters of virtual moment of inertia J and damping coefficient D are shown in Fig. 8.

As shown in Fig. 8, under a load disturbance of 0.05p.u. sudden increase, with the gradual increase of the virtual moment of inertia J, the maximum system frequency deviation decreases slightly from 0.162 Hz to 0.159 Hz with a small reduction amplitude, indicating that increasing the inertia has a certain effect on suppressing frequency drop but the improvement is limited; its main role is reflected in slowing down the rate of frequency change, making the frequency drop process smoother, thereby enhancing the inertial support capability of the system in the initial stage of disturbance resistance. However, a larger J also leads to a slower frequency recovery process and a reduction in the system dynamic response speed. In contrast, the increase of the damping coefficient D improves the frequency regulation performance more significantly. When the damping coefficient D increases gradually, the maximum frequency deviation decreases obviously from 0.144 Hz to 0.104 Hz with a large reduction amplitude, indicating that enhanced damping can effectively suppress the magnitude of frequency drop. Meanwhile, from the perspective of the dynamic process, a larger damping coefficient can weaken the oscillation tendency, make the frequency converge to stability faster, and significantly improve the

Table 1
Main parameters of FSCVSPSU.

Model parameter	Value	Model parameter	Value
Rated head	540 m	Rotational speed	500 r/min
Upstream water level	733-716 m	Downstream water level	181-163 m
Max/Min working head	565/526 m	Rated capacity	306.1 MW
K1	10	*Ch*	0.5
K2	0.8	*Cy*	0.2
Ty	0.2s	*Ta*	8.5s
rs	0.038p.u.	*kp*	0.5
ki	30	*kpi*	9.5
kii	24	*w0*	314 rad · s-1
Udc	25kv	*Ug*	13.8kv

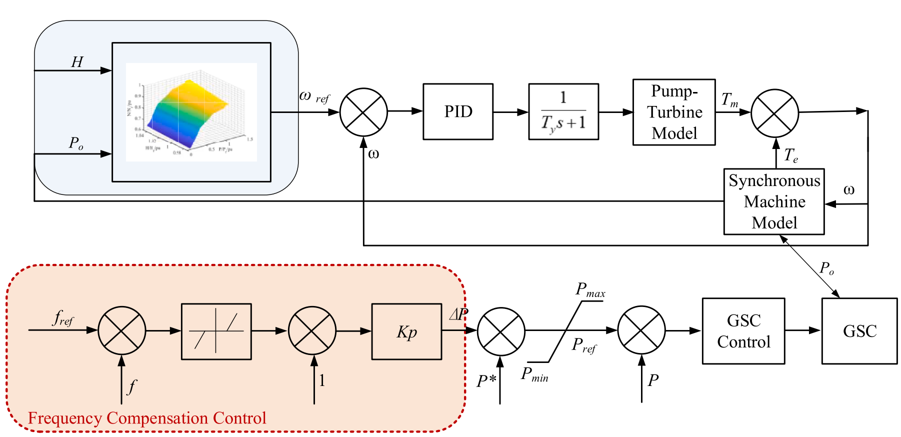

Fig. 7. Control block diagram of a FSCVSPSU participating in primary frequency regulation under the power-priority mode.

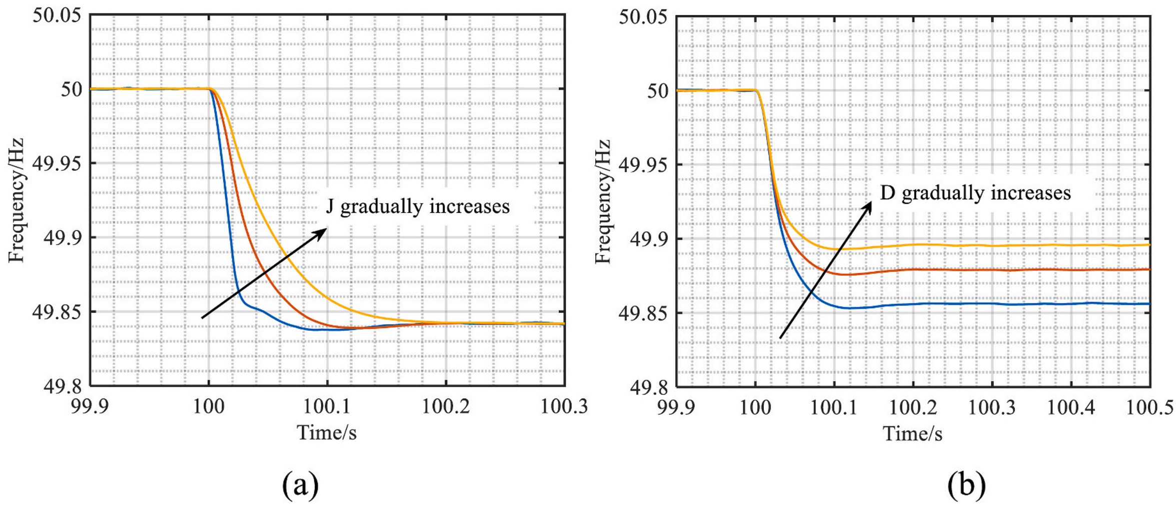

Fig. 8. Frequency responses of the system under different parameters: (a) different virtual moment of inertia J; (b) different damping coefficient D.

dynamic performance and stability of the system.

In this study, a sensitivity analysis is only performed on the virtual moment of inertia J and virtual damping coefficient D. The sensitivity analysis of the parameters in the RBF neural network is not conducted in the current work, which will be further explored and discussed in our future research.

4.1.2. Comparison of multiple regulation control strategies

To verify the effectiveness and superiority of the proposed adaptive VSG control algorithm, a comparative analysis is conducted among four different VSG control strategies. The four strategies considered are:(1) a fixed-parameter VSG control strategy with constant inertia and damping coefficients (J and D); (2) an RBF neural network–based adaptive VIC strategy with adaptive J only; (3) a control strategy with adaptive virtual damping coefficient D only; and (4) the proposed coordinated adaptive control strategy with both virtual inertia and damping coefficients jointly adjusted (*J-D* adaptive).The simulation scenario for the FSCVSPSU under the four control strategies during a load-increase process is defined as follows: the initial upstream and downstream reservoir water levels are set to 729 m and 173 m, respectively. The unit initially operates at 0.6 p.u. of rated power, and a sudden load increase of 0.05 p.u. (relative to rated power) is applied at $t = 50$ s.

As illustrated in Fig. 9, the proposed coordinated adaptive control approach, featuring joint adjustment of virtual inertia and damping coefficients, demonstrates superior dynamic frequency response performance. Specifically, following the load increase, the system frequency undergoes a downward deviation. The fixed-parameter VSG control strategy demonstrates the weakest capability to buffer frequency disturbances. By comparison, the proposed coordinated adaptive strategy enables a more rapid frequency regulation response with mitigated frequency deviation. Specifically, its maximum frequency deviation is 0.0837 Hz, occurring at 50.11 s. This value is significantly lower than those of the fixed-parameter VSG control (0.0982 Hz at 50.12 s) and the RBF neural network-based adaptive VIC strategy (0.0978 Hz at 50.11 s).

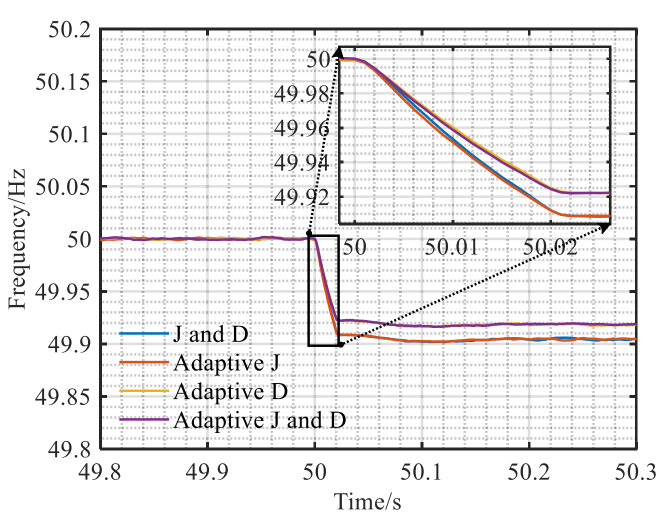

Fig. 9. Time-domain responses of system frequency under different VSG control strategies.

From the perspective of quasi-steady-state performance, the proposed control strategy further reduces the steady-state frequency deviation. Notably, under load disturbances, the coordinated adaptive adjustment of virtual inertia and damping significantly enhances frequency stability and suppresses adverse dynamic frequency deviations. Meanwhile, the system's disturbance rejection capability is improved, along with enhanced dynamic stability. Overall, such coordination remarkably strengthens the frequency stability of the system.

4.1.3. The impact of FCC on frequency regulation characteristics

To elucidate the influence mechanisms of FCC on grid frequency dynamics during frequency regulation, three additional FCC strategies are considered in this study. At $t = 50$ s in the simulation, a sudden load increase of 0.05 p.u. (with respect to rated power) is applied to the system. When the FSCVSPSU engages in grid frequency regulation under various frequency compensation strategies, the dynamic responses of grid frequency and output power are illustrated in Fig. 10, together with those of rotational speed and guide vane opening.

As shown in Fig. 10, the incorporation of different FCC strategies in the FSCVSPSU exerts a pronounced influence on system frequency dynamics and supplementary power response performance. In the absence of FCC, the unit participates in frequency regulation solely through the adaptive VSG strategy. Under this condition, following a sudden load increase, the system exhibits the fastest rate of frequency decline, the largest frequency deviation, and the greatest quasi-steady-state frequency error. Meanwhile, the supplementary power remains unchanged, further confirming that the unit does not actively respond to the frequency disturbance.

When a fixed-parameter FCC is applied, the FSCVSPSU is able to provide a certain level of additional power support during the initial stage of frequency decline, thereby mitigating the rate of frequency reduction and the maximum frequency deviation. Subsequently, the supplementary power gradually converges to a quasi-steady operating point, leading to stabilization of the system frequency. Owing to the

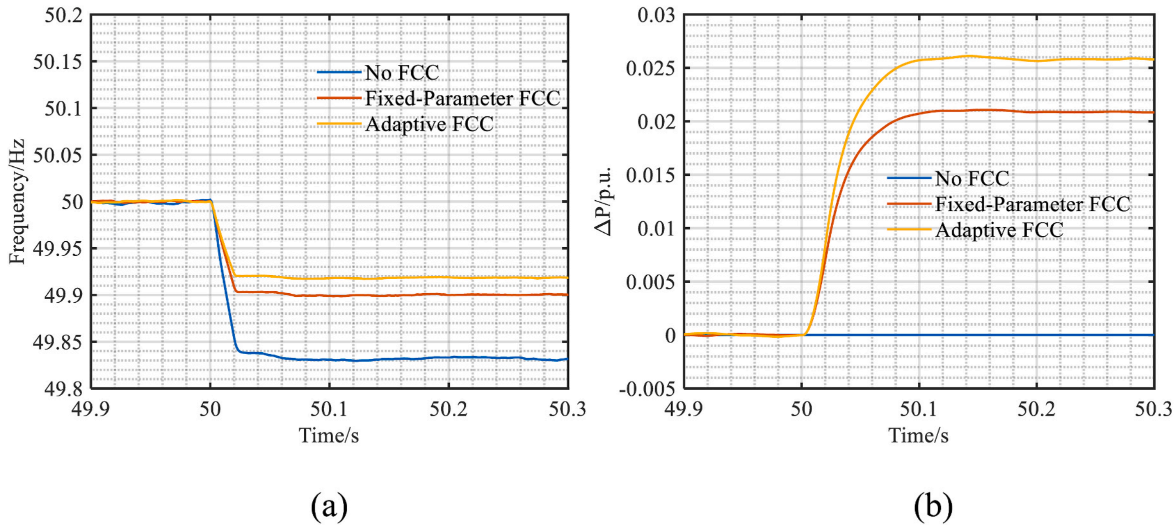

Fig. 10. System dynamic responses of the FSCVSPSU under different FCC strategies: (a) System Frequency; (b) Additional Power Variation of FCC.

sustained output of supplementary power, the steady-state frequency deviation is reduced to some extent.

By contrast, when an adaptive FCC strategy is employed, the FSCVSPSU responds to frequency disturbances in a more rapid and pronounced manner. As illustrated in the figure, the adaptive strategy delivers a larger amount of active power to the grid with a higher ramping rate immediately after the load increase. Consequently, it achieves the smallest frequency deviation, the fastest frequency recovery, and a significantly reduced steady-state frequency error compared with the fixed-parameter compensation strategy. This superior performance can be attributed to the ability of the adaptive control scheme to dynamically adjust the compensation gain according to real-time frequency variations, enabling the unit to promptly enhance supplementary power support and thereby realize improved frequency stability.

In summary, both fixed-parameter and adaptive frequency compensation strategies can significantly enhance the frequency support capability of the system, with the adaptive FCC exhibiting the best overall performance. Specifically, it effectively reduces the maximum frequency deviation, accelerates frequency recovery, and mitigates steady-state frequency error. Therefore, in practical engineering applications, when FSCVSPSUs participate in frequency regulation, the adaptive FCC strategy should be preferentially adopted as the primary frequency regulation control scheme.

4.2. Frequency regulation characteristics of FSCVSPSUs under load increase scenarios

The massive integration of renewable energy sources has notably heightened the volatility and randomness of power systems, thereby augmenting the occurrence frequency of sudden load surge events. This evolution imposes more stringent requirements on frequency regulation resources in terms of response speed, control accuracy, and dynamic adaptability. Owing to their adjustable rotational speed and rapid power regulation capability, FSCVSPSUs represent a highly effective resource for mitigating power imbalances and maintaining grid frequency stability.

Sudden load additions of different magnitudes impose markedly different regulation demands on FSCVSPSUs. During transient frequency regulation processes, the response characteristics of the unit vary accordingly, and these variations ultimately determine the quality and effectiveness of FRP. Therefore, investigating grid load increase events of different capacities is of substantial practical importance for revealing the load adaptability of FSCVSPSUs in frequency regulation and for enhancing the targeting and robustness of frequency control strategies.

For exploring the influencing mechanism of load increment magnitude on the frequency regulation characteristics of FSCVSPSUs, and accurately characterizing their frequency regulation potential under various grid load growth scenarios, this section focuses on analyzing the unit performance under different load-adding conditions. Such an investigation provides a reliable basis for optimizing grid frequency regulation strategies.

In this study, load addition scenarios are adopted as representative cases. Based on the established FSCVSPSU model, the transient frequency regulation processes of the unit under different load addition magnitudes are simulated and analyzed. The simulation conditions are specified as follows: the initial upstream and downstream reservoir water levels are set to 729 m and 173 m, respectively. At the beginning of the simulation, the FSCVSPSU operates in grid-connected generation mode, with power balance maintained between generation and load, and the system frequency remains stable. At $t = 50$ s, incremental loads of 0.05 p.u., 0.10 p.u., and 0.15 p.u. (relative to rated power) are successively applied to an initial load level of 0.9 p.u.

The introduction of additional load disrupts the power balance between generation and demand, thereby inducing system frequency deviations. During the frequency regulation process, the FSCVSPSU operates under a power-priority control mode. The proportional–integral–derivative (PID) parameters of the governor are set to $kp = 3$ and $ki = 0.2$, while the GSC adopts a coordinated control strategy combining adaptive frequency compensation and adaptive VSG control. The dynamic responses of the FSCVSPSU during the frequency regulation process under different load addition scenarios are illustrated in Fig. 11.

From Fig. 11, under load addition disturbances, the FSCVSPSU is capable of providing rapid and effective frequency regulation responses for different load increments (0.05 p.u., 0.10 p.u., and 0.15 p.u.). With increasing load magnitude, the extent of frequency decline, the maximum frequency deviation, and the quasi-steady-state frequency error gradually increase. Nevertheless, the overall frequency response remains well controlled, indicating that the FSCVSPSU possesses strong inertia support capability and primary FRP.

In the early phase of frequency drop, the FSCVSPSU swiftly boosts its active power output in reaction to the frequency deviation, as depicted in Fig. 11(b), all load increments trigger a pronounced rise in supplementary power with a high ramping rate, and the steady-state power output exhibits a positive correlation with the magnitude of the load disturbance. This behavior demonstrates that, through the auxiliary FCC, the GSC of the FSCVSPSU can rapidly regulate its output power to track load variations and provide the required support, thereby

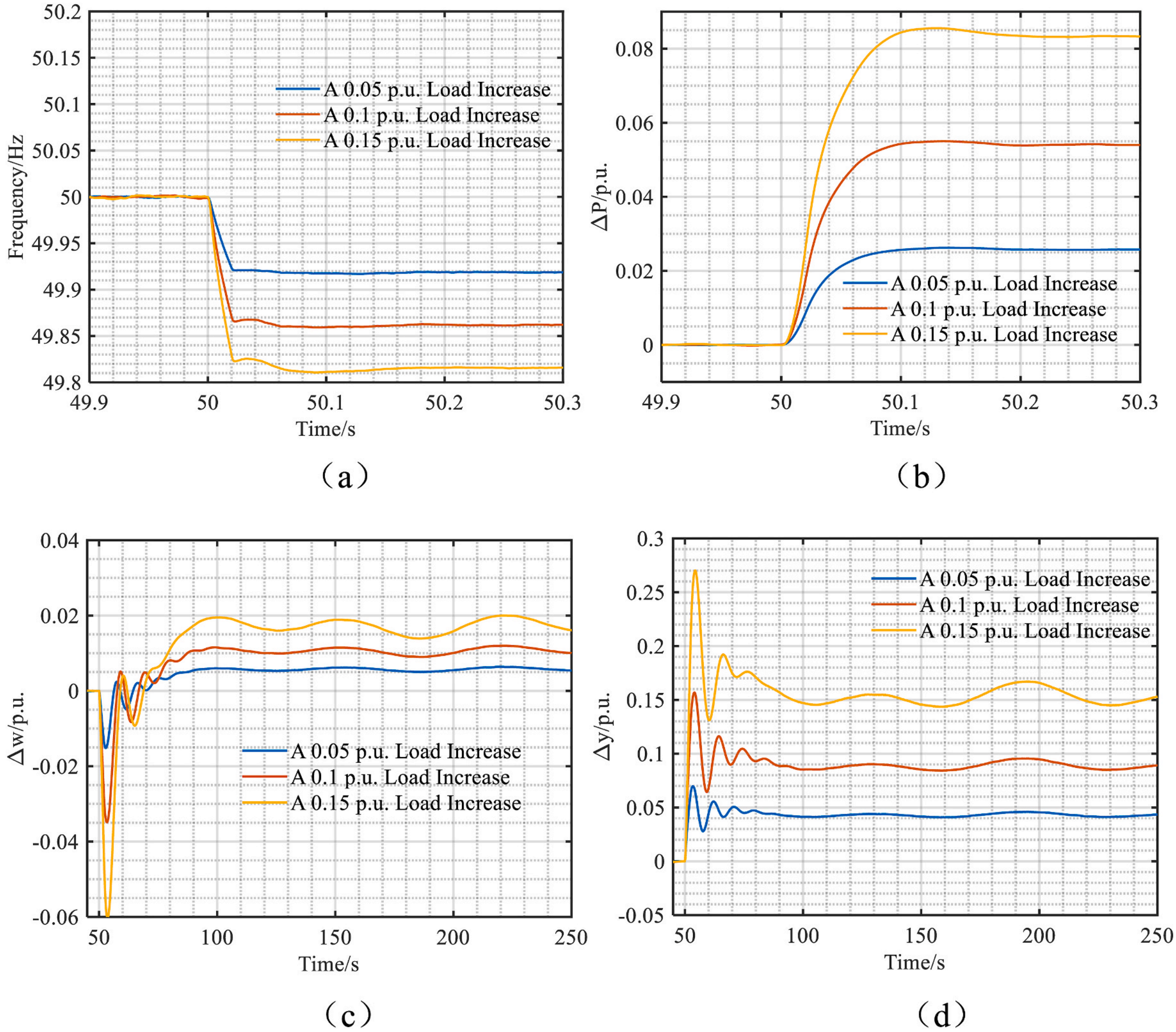

Fig. 11. Dynamic responses of the unit during the frequency regulation process under different load addition scenarios:(a) system frequency; (b) variation of unit supplementary power; (c) variation of unit rotational speed; (d) variation of unit guide vane opening.

mitigating the frequency drop and reducing the maximum frequency deviation.

Fig. 11(c) further shows that load disturbances will cause an instantaneous drop in the rotational speed of the FSCVSPSU, and the speed deviation increases with the load increment. Subsequently, under the action of the governing system, the speed deviation is gradually restored and converges to a new steady state. Although there are minor fluctuations in the speed deviation curve, they are all strictly limited within the allowable range of steady-state error set in this paper, which fully demonstrates that the proposed control strategy can achieve steady-state convergence and stable operation of the system. Under different load conditions, the speed oscillation exhibits a typical damped response, and the oscillation amplitude increases in proportion to the load increment.

The variation in guide vane opening is depicted in Fig. 11(d), which reflects the dynamic adjustment of mechanical power. Following the load disturbance, the guide vane opening increases rapidly and exhibits oscillatory behavior of varying degrees, with both the peak value and regulation time increasing as the load increment becomes larger. The increase in guide vane opening effectively enhances the mechanical input power of the pump-turbine, compensating for the initial speed reduction caused by the sudden increase in electromagnetic power and enabling the unit to gradually reach a new operating equilibrium.

Overall, the FSCVSPSU demonstrates fast electrical response, well-controlled mechanical dynamics, and strong frequency support capability under different load disturbances. Although the maximum frequency deviation, peak speed deviation, and guide vane opening variation all increase with the magnitude of the load disturbance, the system is able to restore a new steady state within a relatively short time under all operating conditions, reflecting high reliability in frequency regulation and robust system stability.

To quantitatively evaluate the FRP of FSCVSPSUs under different load addition scenarios, and to accurately characterize the impact of varying grid load increments on the frequency response of the unit, a set of performance indicators is selected in accordance with established grid codes and frequency stability assessment criteria. According to the Two Detailed Rules for power system operation, pumped storage units subjected to frequency disturbances are required to satisfy technical requirements such as achieving at least 90% of the maximum adjustable power output within 15 s and restoring to a new steady state within 60 s, so as to ensure the timeliness of active power support and overall system stability.

Based on these requirements, the following indicators are adopted for quantitative analysis: power response time (T_p) (defined as the time required for the output power to reach 90% of the load adjustment value), quasi-steady-state frequency deviation (f_{qss}), maximum frequency deviation ($f_{\max}$), steady-state power deviation (P_{ss}), qualified rate of integrated energy (R_{IEQ}), qualified rate of power response (R_{QPR}), maximum rotational speed deviation ($n_{\max}$), and guide vane stroke (D_{gs}). These indicators comprehensively capture the entire dynamic response process of the FSCVSPSU under different load addition magnitudes and enable a clear quantitative comparison of performance differences. The corresponding statistical and computational results are summarized in Table 2, providing a reliable basis for control parameter optimization and the formulation of differentiated frequency regulation strategies.

The calculated results presented in Table 2 indicate that, as the magnitude of the load disturbance gradually increases, the FSCVSPSU exhibits a series of systematic and discernible trends in both frequency and power regulation processes, accompanied by a continuous escalation of overall frequency regulation stress on the system.

First, both the quasi-steady-state frequency deviation and the peak frequency deviation increase significantly with the elevation of disturbance magnitude, in terms of FRP. This behavior clearly indicates that under larger power shocks, the system experiences deeper transient frequency drops and requires a longer duration to recover to a new steady-state operating point. The concurrent increase in steady-state power deviation further reflects that, when responding to larger disturbances, the unit must provide greater frequency regulation output, thereby resulting in a larger deviation of the final operating point from the initial condition. Collectively, these phenomena demonstrate that under large-scale load disturbance scenarios, grid frequency stability becomes increasingly dependent on the regulation capability of the FSCVSPSU, while the regulation burden borne by the unit itself rises substantially.

From the perspective of frequency regulation contribution, both the qualified rate of integrated energy and the qualified rate of power response exhibit an upward trend with increasing disturbance magnitude. This indicates that the FSCVSPSU assumes a greater share of frequency support and power compensation under more severe disturbances, leading to a significant enhancement in regulation intensity. This trend is closely related to the unit's regulation mechanism: when subjected to larger disturbances, the FSCVSPSU must rapidly increase the power output of the grid-side FSC to suppress frequency decline and facilitate system recovery toward a new equilibrium. Therefore, the increase in frequency regulation contribution is both a necessary response to intensified disturbances and a manifestation of the unit's enhanced emergency regulation capability.

Compared with frequency- and power-related indicators, the mechanical-side dynamic responses exhibit higher sensitivity to load disturbances. As shown in Table 2, both the maximum rotational speed deviation and the guide vane stroke increase progressively with disturbance magnitude. This indicates that, under larger disturbances, the FSCVSPSU is required not only to withstand higher electromagnetic regulation stress but also to undergo more pronounced mechanical dynamic variations during frequency regulation. In particular, the significant expansion of guide vane stroke implies that the actuator system must execute larger and faster adjustments within a short time span, which may lead to cumulative mechanical fatigue, aggravated wear, and intensified local hydraulic excitation. An increase in rotational speed deviation further indicates that the mismatch between electromagnetic and mechanical torques becomes more prominent during the initial disturbance phase, subjecting the unit to stronger inertial impacts. This phenomenon, which stems from the imbalanced state of the two torques, may pose potential threats to shafting stability and operational reliability.

In summary, as the load disturbance magnitude increases, the pressure on the FSCVSPSU to maintain system frequency stability intensifies significantly. The amplification of frequency deviations, power deviations, and mechanical dynamic fluctuations collectively indicates that the system faces increasingly severe stability challenges under large-disturbance scenarios. Therefore, in practical engineering applications, it is necessary to further optimize frequency regulation control strategies to enhance the coordination between electromagnetic and mechanical regulation mechanisms. Meanwhile, the integration of diversified frequency regulation resources—such as energy storage systems and fast-response generating units—should be considered to jointly share the regulation burden under high-disturbance conditions. Such measures can effectively preserve the stability of power grid frequency while alleviating dynamic mechanical stresses on the unit. Notably, this facilitates the long-term safe and stable operation of both the FSCVSPSU and the overall power system.

4.3. Frequency regulation characteristics of FSCVSPSUs under load shedding scenarios

The previous section studied the impact on the frequency regulation process under the load input scenario. However, in the actual operation of the power system, load shedding is also a typical and non-negligible disturbance form, which will cause instantaneous power surplus in the system and lead to a rapid increase in frequency, putting higher requirements on the downward regulation capacity and frequency suppression performance of the unit. Therefore, it is necessary to further analyze the frequency regulation characteristics under the load shedding scenario. The operating conditions are set such that at $t = 50$s, the load is shed by 0.05p.u. rated power, 0.1p.u. rated power, and 0.15p.u. rated power respectively from 0.9p.u. rated power. The dynamic response of the frequency regulation process of the FSCVSPSU is shown in Fig. 12.

Fig. 12 shows the dynamic response of the unit during the frequency regulation process under different load shedding scenarios. It can be seen from Fig. 12(a) that the instantaneous load shedding leads to a rapid increase in frequency, and the maximum frequency deviation increases monotonically with the increase of disturbance amplitude. Among them, the frequency peak is the highest under the 0.15p.u. working condition, reaching 50.19 Hz. However, all working conditions can recover to a new steady state in a short time, and the process is smooth without continuous oscillation or secondary overshoot, indicating that the system has good damping characteristics and stability. At the same time, Fig. 12(b) shows that the unit's compensation power decreases rapidly, and its variation range is basically proportional to the load shedding amount. Even under large disturbances, there is no obvious oscillation or over-regulation, reflecting a strong power down-regulation capability. Fig. 12(c) shows that the unit speed experiences a short-term acceleration accompanied by small oscillations in the initial stage of the disturbance, and then quickly decays to a stable value. It can be seen from Fig. 12(d) that the guide vane opening decreases rapidly in the early stage of frequency regulation to reduce the hydraulic output power, and its regulation range increases with the enhancement of the disturbance. During the frequency regulation process, the guide vane opening presents oscillation attenuation and finally reaches a stable state. In general, load shedding is a typical power surplus disturbance, and its frequency regulation process is essentially a process of rapid power reduction response of the unit. The state variables are coordinated consistently, and the system response is mainly characterized by amplitude amplification with the increase of disturbance amplitude while the dynamic characteristics remain basically unchanged, which

Table 2
Calculated FRP metrics under different load addition scenarios.

Condition	T_p/s	f_{qss}/Hz	$f_{\max}$/Hz	P_{ss}/p.u.	R_{IEQ}/p.u.	R_{QPR}/p.u.	$n_{\max}$/p.u.	D_{gs}/p.u.
0.05p.u.	0.23	0.081	0.083	0.05	0.899	1	0.021	0.182
0.1p.u.	0.23	0.13	0.14	0.1	1	1	0.05	0.37
0.15p.u.	0.23	0.184	0.189	0.15	1	1	0.08	0.53

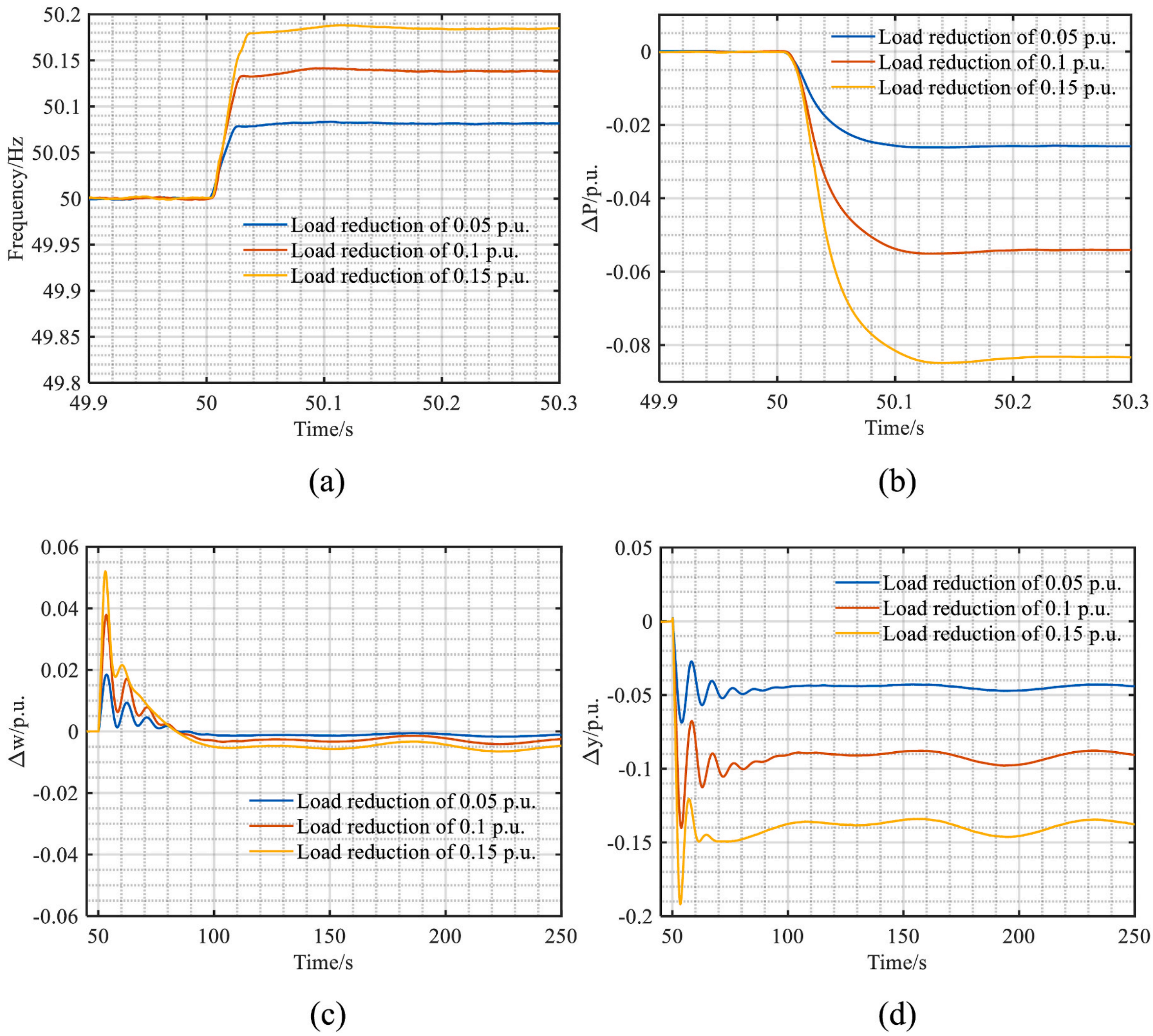

Fig. 12. Dynamic responses of the unit during the frequency regulation process under different load reduction conditions:(a) system frequency; (b) variation of unit supplementary power; (c) variation of unit rotational speed; (d) variation of unit guide vane opening.

verifies the robustness and effectiveness of the proposed control strategy under such working conditions.

This section further calculates and quantitatively evaluates the frequency regulation performance of the FSCVSPSU under different load shedding operating scenarios, and the results are shown in Table 3.

It can be seen from Table 3 that under different load shedding conditions, the unit power response time slightly increases from 0.229s to 0.24s with a small overall variation range, indicating that the unit can still maintain fast and stable response capability when the disturbance is enhanced, and the dynamic performance is limitedly affected by the disturbance amplitude. Secondly, both the quasi-steady-state frequency deviation and the maximum frequency deviation increase significantly with the increase of load shedding, rising from 0.082 Hz to 0.083 Hz to 0.185 Hz and 0.19 Hz respectively, which shows that the system has weak overshoot phenomenon, stable frequency response process, and the control strategy has good suppression ability for frequency fluctuations. The steady-state power deviation is basically consistent with the disturbance amplitude, reflecting that the unit can accurately track the frequency modulation command and realize effective power distribution; the qualified rate of integral electric quantity increases from 0.9 to 1, and the qualified rate of output response remains 1 all the time, indicating that the unit's regulation accuracy and execution performance are further improved under large disturbances, meeting the frequency modulation assessment requirements. In addition, the maximum unit speed deviation increases from 0.02 p.u. to 0.06 p.u., and the guide vane stroke expands from 0.22 p.u. to 0.40 p.u., both increasing with the increase of disturbance amplitude, but the overall value is still within a reasonable range, indicating that the unit has good mechanical adaptability and actuator coordination.

Table 3
FRP indicators calculated under different removal loads.

Condition	T_p/s	f_{qss}/Hz	$f_{\max}$/Hz	P_{ss}/p.u.	R_{IEQ}/p.u.	R_{QPR}/p.u.	$n_{\max}$/p.u.	D_{gs}/p.u.
0.05p.u..	0.229	0.082	0.083	0.05	0.9	1	0.02	0.22
0.1p.u.	0.233	0.14	0.14	0.1	1	1	0.04	0.35
0.15p.u.	0.24	0.185	0.19	0.15	1	1	0.06	0.4

4.4. Frequency regulation characteristics of FSCVSPSUs under different hydraulic head conditions

Operating hydraulic head is a key characteristic parameter that reflects water flow inertia, and its variation directly influences the response behavior of the hydraulic system. More importantly, the speed optimization system of FSCVSPSUs takes power and hydraulic head as core input variables. Consequently, the optimal rotational speed varies with different head conditions. Such variations further affect the hydraulic operating characteristics of the unit, as well as its FRP. Therefore, investigating different hydraulic head scenarios is of significant value for clarifying the coupling mechanisms among hydraulic head, speed optimization, and frequency regulation characteristics, as well as for enhancing the engineering applicability of FSCVSPSUs.

To reveal the influence mechanism of hydraulic head variations on the FRP of FSCVSPSUs, this study considers a series of head conditions ranging from 0.99 p.u. to 1.02 p.u. of the rated head, with an interval of 0.01 p.u. For each scenario, a sudden load increase of 0.05 p.u. (relative to rated power) is applied at $t = 50$ s, starting from an initial operating point of 0.9 p.u. rated power. The transient frequency regulation processes of the FSCVSPSU under different hydraulic head conditions are simulated, and the corresponding dynamic response results are illustrated in Fig. 13.

From Fig. 13, under different operating hydraulic head conditions, the FSCVSPSU exhibits nearly identical frequency and supplementary power regulation performance when subjected to the same load disturbance. The system frequency trajectories and supplementary active power responses almost overlap across all head scenarios, indicating that the frequency regulation quality of the FSCVSPSU is highly independent of hydraulic head variations. In other words, the hydraulic head parameter is effectively decoupled from system frequency dynamics.

However, notable differences can be observed in the rotational speed deviation and guide vane opening responses. As the operating head decreases, the oscillation amplitudes following the disturbance become more pronounced. In particular, during the speed droop stage and the initial guide vane response, low-head conditions (e.g., 0.99 p.u.) exhibit more evident peak–valley fluctuations, with oscillation magnitudes slightly larger than those under higher head conditions. The underlying reason lies in the weaker water flow kinetic energy associated with a lower hydraulic head, which restricts the pump-turbine's capacity to promptly deliver mechanical support amid rapid electromagnetic power fluctuations. Consequently, the torque imbalance in the initial disturbance stage becomes more significant, leading to intensified dynamic responses in rotational speed and guide vane motion. By comparison, higher water head mitigates water flow inertia, optimizes torque matching, and yields a more stable mechanical response.

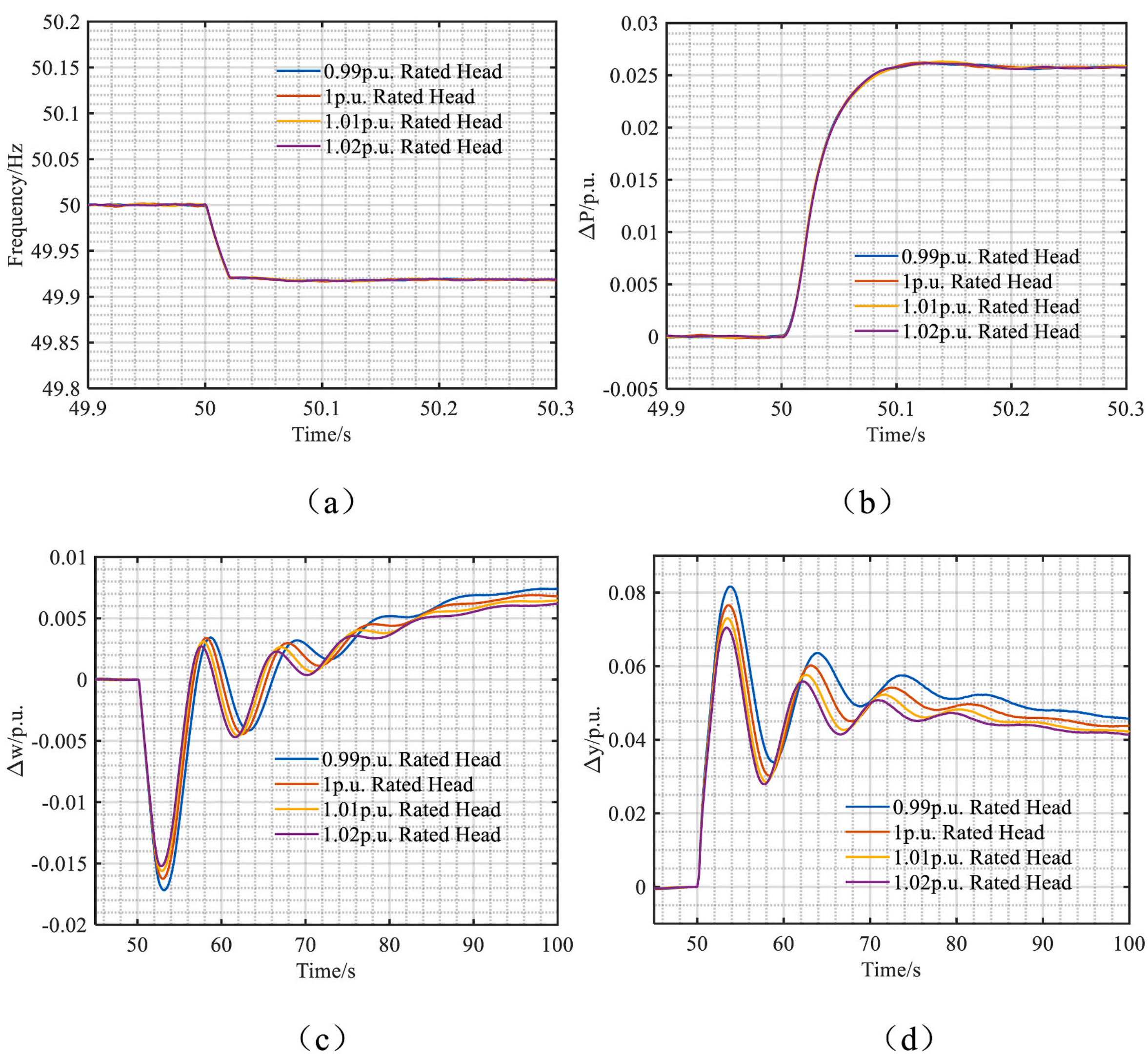

Fig. 13. Time-domain dynamic responses of the FSCVSPSU during the frequency regulation process under load disturbance at different hydraulic head conditions:(a) system frequency; (b) variation of unit supplementary power; (c) variation of unit rotational speed; (d) variation of unit guide vane opening.

Although higher hydraulic head is beneficial for suppressing oscillations and enhancing regulation stability, the increased kinetic energy of the water flow may also introduce potential risks, such as intensified hydraulic transients and increased forces acting on the guide vanes. Under frequent disturbances, these effects may accelerate fatigue accumulation and impose additional stress on hydraulic components. Therefore, while hydraulic head variations have negligible influence on the frequency and power regulation indicators of FSCVSPSUs, they can substantially alter mechanical dynamic behavior, thereby affecting operational smoothness and structural loading levels.

Based on these observations, it is recommended that enhanced monitoring of hydraulic component loading be implemented under high-head operating conditions, while under low-head conditions, appropriate optimization of mechanical torque compensation or guide vane control strategies should be considered. Such measures can help improve mechanical safety and operational reliability without compromising FRP.

Building upon the definitions of FRP indicators presented in the previous section, a systematic calculation and quantitative evaluation of the FRP of FSCVSPSUs under different hydraulic head scenarios are further conducted in this study, with the results summarized in Table 4. To comprehensively capture the impact of hydraulic head variations on regulation performance, the table includes indicators related to frequency dynamics and steady-state deviations, power response characteristics, frequency regulation contribution metrics, as well as key mechanical dynamic variables such as rotational speed and guide vane behavior. By comparing these indicators across different head conditions, the influence mechanisms of hydraulic head variations on the regulation process can be examined in depth, providing reliable data support for assessing frequency support capability and operational stability of FSCVSPSUs under multi-head operating environments.

The data reported in Table 4 indicate that, under different operating conditions, all frequency- and power-related regulation indices exhibit only marginal variations. Specifically, the quasi-steady-state frequency deviation remains within a narrow range of 0.0816–0.0821 Hz, while the maximum frequency deviation is consistently maintained at approximately 0.083–0.084 Hz. The steady-state power deviation remains constant at 0.05 p.u., and both the qualified rate of integrated energy and the qualified rate of power response are sustained within stable intervals of 0.89–0.90 and approximately 1 p.u., respectively. Meanwhile, the maximum rotational speed deviation and guide vane stroke also vary within very limited ranges, fluctuating only slightly between 0.021 and 0.025 p.u. and 0.18–0.20 p.u., respectively.

These results demonstrate that variations in operating hydraulic head do not exert a significant influence on the FRP or power response characteristics of the FSCVSPSU. The key regulation indices exhibit a high degree of consistency and stability across all considered head conditions. This behavior can be primarily attributed to the control architecture of the FSCVSPSU, in which frequency regulation is predominantly governed by active power modulation on the grid-side FSC. Consequently, the FRP exhibits weak sensitivity to variations in hydraulic head, thereby achieving effective decoupling of frequency dynamics from mechanical operating conditions.

Nevertheless, despite the overall stability of the regulation indices, slight fluctuations in guide vane stroke and rotational speed deviation can still be observed in Table 4. This suggests that under larger disturbances or more extreme operating conditions, the mechanical components of the unit may be exposed to additional stress accumulation or localized amplification of dynamic responses, which warrants attention in long-term operation. Overall, the FSCVSPSU is able to maintain stable frequency and power regulation characteristics under the investigated conditions, and its regulation performance remains largely insensitive to variations in operating hydraulic head, thereby demonstrating strong robustness.

4.5. Frequency regulation characteristics of FSCVSPSUs under different initial power conditions

The initial operating power serves as a prerequisite operating condition for unit participation in frequency regulation, and its suitability directly affects the quality of FRP. To date, the frequency regulation characteristics of FSCVSPSUs under multiple initial operating power conditions have not been systematically investigated. To address this research gap, this study considers a rated hydraulic head condition of 1.01 p.u. and selects five initial operating power levels- 0.7 p.u., 0.8 p.u., and 0.9 p.u. of rated power-as representative operating scenarios. Based on these conditions, the transient frequency regulation processes of the FSCVSPSU under different initial power levels are simulated. The corresponding dynamic response results are presented in Fig. 14.

As illustrated by the response curves under different initial power levels in Fig. 14, the FSCVSPSU exhibits highly consistent dynamic characteristics in both system frequency and active power regulation following a load disturbance. This observation is consistent with the conclusions drawn for different hydraulic head scenarios. Regardless of the initial operating point, the magnitude of frequency deviation, the transient recovery process, and the final steady-state value are almost identical. Similarly, the shape of the supplementary power ramping curves, their peak values, and the time required to reach steady state remain essentially unchanged.

These results indicate that the dominant factor governing the frequency regulation behavior of the FSCVSPSU is the rapid active power modulation on the grid-side FSC, rather than the initial operating power or hydraulic operating condition. Consequently, the FRP is primarily determined by the vector control loops and their dynamic response mechanisms, leading to highly consistent regulation quality across different initial power levels.

In contrast to the strong overlap observed in frequency and power responses, the dynamic behaviors of rotational speed deviation and guide vane stroke exhibit a certain degree of differentiation. Under lower initial power conditions (e.g., 0.7 p.u.), the unit experiences the smallest speed droop but the largest speed oscillation amplitude, accompanied by increased peak guide vane opening and more pronounced oscillations. By comparison, the case with an initial power of 0.8 p.u. demonstrates relatively smoother and more stable dynamic behavior. This trend arises not only from the significant difference in response speed between electromagnetic power and mechanical torque following a disturbance, but also from variations in the optimal speed reference of the FSCVSPSU during transient regulation.

Specifically, mechanical torque recovery is inherently constrained by water flow inertia and hydraulic energy conversion processes, causing it to lag behind the rapid adjustment of electromagnetic power. As a result, a more pronounced torque imbalance emerges in the initial stage of the disturbance, which excites larger speed oscillations. Meanwhile, the deviation between the optimal speed reference and the actual rotational

Table 4
Calculated FRP metrics under different operating hydraulic head conditions.

Condition	T_p/s	f_{qss}/Hz	$f_{\max}$/Hz	P_{ss}/p.u.	R_{IEQ}/p.u.	R_{QPR}/p.u.	$n_{\max}$/p.u.	D_{gs}/p.u.
0.99p.u.	0.225	0.0816	0.083	0.05	0.9	1	0.025	0.2
1p.u.	0.226	0.0817	0.0834	0.05	0.89	1	0.023	0.19
1.01p.u.	0.226	0.082	0.083	0.05	0.89	1	0.0221	0.1887
1.02p.u.	0.225	0.0819	0.0832	0.05	0.896	1	0.021	0.1816

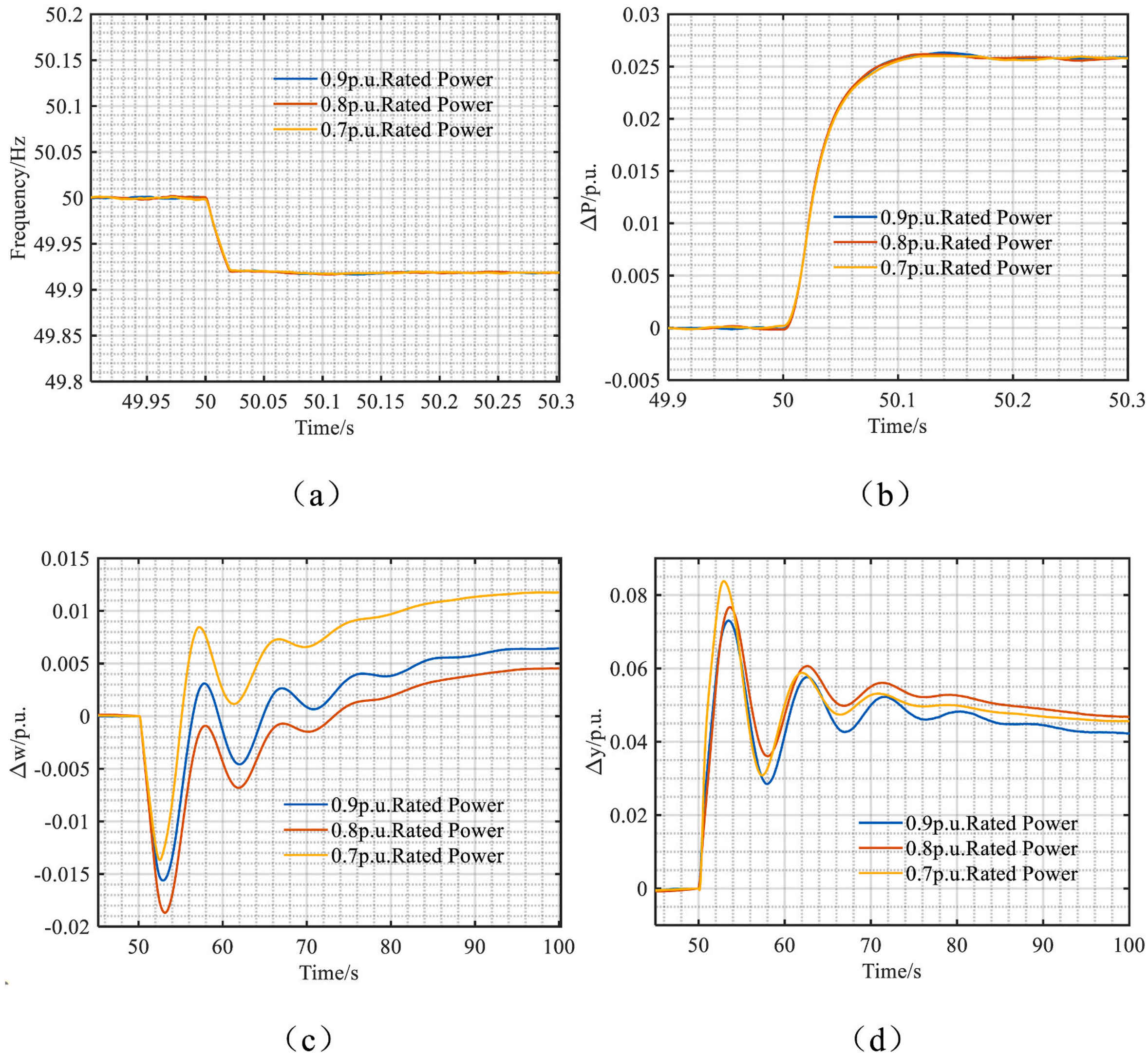

Fig. 14. Time-domain dynamic responses of the FSCVSPSU during the frequency regulation process under load disturbance at different initial power levels:(a) system frequency; (b) variation of unit supplementary power; (c) variation of unit rotational speed; (d) variation of unit guide vane opening.

speed directly determines the magnitude of guide vane adjustment and further influences the dynamic evolution of mechanical torque. When this deviation increases, the guide vane opening expands rapidly to enhance mechanical torque, thereby more effectively compensating for the electromagnetic torque variation and enabling faster torque matching and dynamic stabilization of the unit.

Building upon the FRP index definitions presented in Section 4.2, this study further performs systematic calculation and quantitative assessment of the FSCVSPSU's FRP under various initial power conditions, with the corresponding results compiled in Table 5. To ensure a comprehensive and accurate assessment, the table includes indicators related to frequency deviation, power response characteristics, frequency regulation contribution, as well as key mechanical dynamic variables such as rotational speed and guide vane behavior. By comparing these indicators across different initial power conditions, the influence of initial operating power on frequency regulation quality and dynamic behavior can be examined in depth.

The calculated results presented in Table 5 indicate that the FRP of the FSCVSPSU exhibits a high degree of consistency and stability under different initial operating power conditions. First, with respect to frequency regulation indices, the power response time remains within an extremely narrow range of 0.226–0.227 s, while the quasi-steady-state frequency deviation varies only slightly between 0.0811 Hz and 0.0820 Hz. Similarly, the maximum frequency deviation is stably maintained within 0.0832–0.0845 Hz. These results demonstrate that the unit possesses a very fast power response capability and stable frequency recovery characteristics following disturbances, indicating strong robustness of FRP with respect to variations in operating conditions.

Further examination of the steady-state power deviation shows that

Table 5
Calculated FRP metrics under different initial operating power conditions.

Condition	T_p/s	f_{qss}/Hz	$f_{\max}$/Hz	P_{ss}/p.u.	R_{IEQ}/p.u.	R_{QPR}/p.u.	$n_{\max}$/p.u.	D_{gs}/p.u.
0.9p.u.	0.226	0.0819	0.0834	0.05	0.89	1	0.0221	0.1887
0.8p.u.	0.226	0.0811	0.0835	0.05	0.8952	1	0.023	0.17
0.7p.u.	0.226	0.082	0.0832	0.05	0.8977	1	0.0254	0.1914

it remains constant at 0.05 p.u. across all operating scenarios, confirming that the power matching process after load disturbances is highly stable and that the unit can accurately reach a new steady-state equilibrium. In addition, the qualified rate of integrated energy and the qualified rate of power response are maintained within the ranges of 0.89-0.8977 p.u. and 1 p.u., respectively. Although slight variations exist, the overall stability of these indices remains excellent. Collectively, these results demonstrate that, under the selected disturbance conditions and control strategy, the FRP of the FSCVSPSU is almost insensitive to changes in the initial operating state.

By comparison, the mechanical-side dynamic indices exhibit relatively more pronounced, though still limited, variations. The maximum rotational speed deviation fluctuates within the range of 0.0204–0.0254 p.u., while the guide vane stroke varies between 0.1549 and 0.1914 p.u. Although these differences are insufficient to significantly affect FRP, they reflect the sensitivity of the mechanical system to operating conditions. This behavior can be explained by the fact that frequency regulation of the FSCVSPSU primarily relies on the rapid adjustment of GSC output power, which is completed almost instantaneously, and the electromagnetic power response of the synchronous generator is likewise immediate. In contrast, mechanical torque adjustment is constrained by water flow inertia, hydraulic passage characteristics, and guide vane actuation dynamics, resulting in a noticeably slower response than that of electromagnetic power.

Consequently, following a disturbance, a transient torque imbalance arises between the rapidly varying electromagnetic torque and the more slowly responding mechanical torque, leading to slight differences in rotational speed deviation and guide vane motion under different operating conditions. In summary, the results in Table 5 confirm that the FSCVSPSU exhibits highly stable frequency and power regulation capabilities, with regulation performance that is largely independent of operating conditions, thereby demonstrating strong robustness on the electrical side. Nevertheless, mechanical dynamic parameters still exhibit minor variations with changing operating conditions, which may introduce cumulative stress risks over long-term operation and therefore warrant further attention in practical applications.

4.6. Hydraulic dynamic characteristics of FSCVSPSUs under different operating scenarios

The preceding analysis has examined the dynamic characteristics of system frequency and unit output power. However, it should be noted that during the frequency regulation transient, strong coupling exists between the hydraulic and mechanical subsystems. In the frequency regulation process of FSCVSPSUs, significant oscillations in rotational speed may occur, which in turn can induce pronounced pressure fluctuations in the hydraulic system. The associated hydraulic pressure risks therefore warrant particular attention.

Accordingly, to investigate the hydraulic system behavior of FSCVSPSUs during frequency regulation, all previously considered operating scenarios are adopted in this section, with a specific focus on load increase disturbances as a representative frequency regulation condition. Based on numerical simulations, the maximum spiral casing pressure and the minimum draft tube pressure of the FSCVSPSU under these scenarios are obtained, as illustrated in Fig. 15.

From the three-dimensional distributions of the maximum spiral casing pressure and the minimum draft tube pressure shown in Fig. 15, it can be observed that variations in operating hydraulic head and initial operating power exert a pronounced influence on the internal pressure distribution of the unit. In particular, the maximum spiral casing pressure exhibits a clear and monotonic increasing trend with increasing hydraulic head, whereas the minimum draft tube pressure decreases progressively as the head rises, forming a distinct inverse relationship.

Specifically, under identical initial power conditions, when the operating head increases from 0.99 p.u. to 1.02 p.u., the maximum spiral casing pressure increases by approximately 18–24 m, while the minimum draft tube pressure decreases by about 0.5–2 m. Cross-comparison among different initial power levels further reveals that the spiral casing pressure tends to be higher under lower initial power conditions, while the minimum draft tube pressure becomes correspondingly lower. This indicates that at lower operating power, the reduced water flow kinetic energy amplifies the effect of head variation on pressure distribution.

The underlying mechanisms can be explained as follows. On the one hand, a higher operating head increases the hydraulic energy entering the flow passages, leading to a substantial rise in the pressure head within the spiral casing region. On the other hand, high-head conditions facilitate the deepening of negative pressure zones in the draft tube, resulting in a reduction of the minimum pressure. Moreover, under low initial power operation, the reduced discharge leads to a less uniform distribution of hydraulic energy, further intensifying pressure gradients under high-head conditions and making both the spiral casing and draft tube pressures more sensitive to head variations.

Although the observed pressure variation patterns are consistent with the inherent hydraulic characteristics of pump-turbine systems, they also imply potential structural and hydraulic stability risks for FSCVSPSUs under certain operating conditions. On the one hand, the continuous increase in spiral casing pressure subjects pressure-bearing components to elevated mechanical loads, which, if sustained under high-head operation, may accelerate local structural fatigue and adversely affect equipment lifetime. On the other hand, the reduction in minimum draft tube pressure brings it closer to the cavitation threshold. Under combined high-head and low-initial-power conditions, cavitation is more likely to be triggered, potentially leading to intensified draft tube vibration, cavitation noise, and other adverse phenomena that threaten hydraulic stability.

Overall, the coupled effects of operating hydraulic head and initial

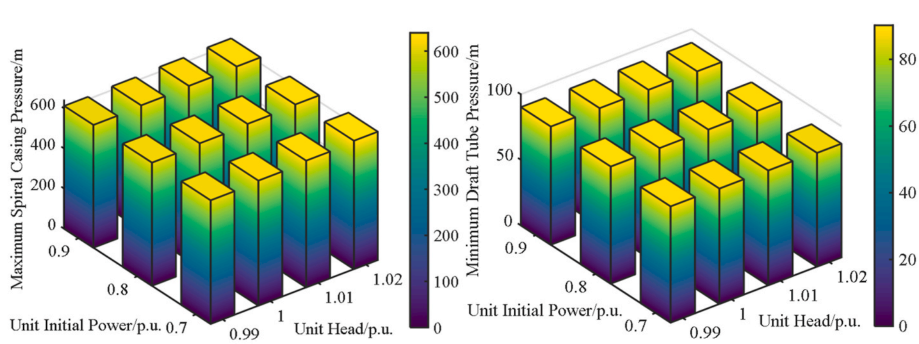

Fig. 15. Statistical results of the maximum spiral casing pressure and the minimum draft tube pressure of the unit during frequency regulation under load disturbance conditions:(a) spiral casing; (b) draft tube.

power have a significant impact on the internal pressure distribution of FSCVSPSUs. High-head and low-power operating conditions can jointly result in elevated spiral casing pressure and deepened draft tube negative pressure, thereby increasing the risks of hydraulic shock and cavitation. Accordingly, in practical engineering applications, it is recommended to reasonably limit the operating head range and avoid prolonged operation under excessively high head conditions. In addition, hydraulic risks can be mitigated by optimizing guide vane control strategies, improving flow passage design, and strengthening online monitoring of pressures in the spiral casing and draft tube, so as to ensure long-term, safe, and stable operation of pumped storage units across the full operating envelope.

5. Conclusions

To enhance the frequency support capability of FSCVSPSUs, this study proposes a coordinated control strategy that integrates an adaptive VSG with frequency compensation. In the proposed framework, a RBF neural network is employed to realize online adaptive regulation of virtual inertia, while an adaptive damping adjustment scheme based on frequency deviation is introduced to improve dynamic stability under islanded and weak-grid conditions. In addition, a frequency compensation loop is incorporated to strengthen the power response under fast disturbances, thereby achieving a favorable balance between frequency regulation depth and dynamic performance. Based on this control architecture, the frequency regulation characteristics of FSCVSPSUs under different load disturbances, operating hydraulic heads, and initial power conditions are systematically analyzed. Through a comprehensive comparison of frequency, power, rotational speed, guide vane motion, and hydraulic pressure indicators, the regulation mechanism and performance advantages of FSCVSPSUs are revealed, providing both theoretical insights and engineering guidance for high-quality frequency control design. The main conclusions are summarized as follows:

Coordinated adaptive control, as proposed herein, integrates adaptive VSG and frequency compensation. This strategy simultaneously enhances the electrical-side inertia support and primary frequency regulation capability of FSCVSPSUs. This strategy effectively improves the rapid response of the unit to frequency disturbances and significantly enhances overall system dynamic stability, offering a practical and reliable solution for improving frequency stability in islanded and weak-grid environments under the context of modern power systems.

Simulation results under different load increments, operating hydraulic heads, and initial power levels demonstrate that FSCVSPSUs exhibit robust FRP. Although increasing disturbance magnitude leads to concurrent increases in maximum frequency deviation and quasi-steady-state frequency error, the combined action of frequency compensation and fast power regulation allows the unit to rapidly and effectively suppress frequency deviations, reflecting strong inertia support and excellent primary frequency regulation characteristics. Variations in hydraulic head and initial operating power have only a limited impact on frequency response and power recovery processes, indicating a high degree of decoupling between frequency regulation quality and operating conditions. In contrast, mechanical-side dynamics are more sensitive to operating scenarios: under low-head, low-initial-power, and large-disturbance conditions, rotational speed oscillations, guide vane stroke, and internal pressure fluctuations are amplified, potentially increasing the risks of hydraulic shock, cavitation, and component fatigue. These findings highlight the necessity of coordinated consideration of electrical regulation performance and mechanical safety in practical applications.

The research conclusions of this paper have been verified through simulation experiments based on actual power station parameters. However, due to the limitations of the objective conditions of current engineering applications and the availability of measured data of FSCVSPSUs, on-site test verification has not been carried out in this study. To further improve the engineering verification level of the research results, a hardware-in-the-loop (HIL) simulation platform for the frequency regulation control of FSCVSPSUs will be built in the future, and systematic HIL experimental verification will be carried out. Through this platform, the dynamic response performance and engineering practical feasibility of the proposed control strategy will be thoroughly tested, providing more sufficient technical support for subsequent on-site engineering applications. In addition, this study only conducted sensitivity analysis on the virtual moment of inertia J and virtual damping coefficient D, and did not carry out sensitivity analysis on the relevant parameters of the RBF neural network, which will be further in-depth explored and improved in subsequent research. At the same time, the depth of analysis on the impact of water head in this study is still insufficient. In the future, we will focus on improving the research on the mechanism of water head impact, and supplement the relevant analysis under variable disturbances and multiple operating states of the system, so as to enhance the comprehensiveness and reliability of the research results.

CRediT authorship contribution statement

Zhang Liu: Writing – original draft, Methodology, Conceptualization. **Diyi Chen:** Writing – review & editing, Methodology, Funding acquisition. **Yunpeng Zhang:** Visualization, Validation. **Tiantian Wang:** Visualization, Data curation. **Ziwen Zhao:** Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the following grants: National Natural Science Foundation of China (Grant No.: 52409120), National Natural Science Foundation of China (Grant No.: 52339006),Free exploration basic research (Grant No.: 2024ZY-JCYJ-02-37). International Science & Technology Cooperation Program of the 2025 Autonomous Region Key R&D and Achievement Transformation Plan (Science & Technology Cooperation) (Grant No.: 2025KJHZ0043).

Data availability

The data that has been used is confidential.

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